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RELIABLE HIGH-SPEED, HIGH-DENSITY CONNECTOR AND INTERCONNECTION SYSTEM THEREOF

Abstract

Reliable high-speed, high-density connectors and interconnection systems thereof. An interconnection system includes a plug connector and a receptacle connector. The plug connector comprises a housing assembly, a module, and a CPA device. The module includes densely disposed conductive pads. The module can include a circuit board with the conductive pads and high-speed data cables having ends mounted onto the conductive pads. The CPA device is disposed on the housing assembly, and configured for securing the plug connector to the receptacle connector. The CPA device can include locking features disposed on different sides of the housing assembly, which can improve reliability. The receptacle connector includes an outer housing having complementary locking features. The connectors can be electrically connected as long as a first locking feature in its locked position. Such techniques can enable interconnection systems for harsh environments such as one presented by an automobile.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority to and the benefit of Chinese Patent Application No. 202420469174.1, filed on Mar. 11, 2024. This application also claims priority to and the benefit of Chinese Patent Application No. 202410277511.1, filed on Mar. 11, 2024. The contents of these applications are incorporated herein by reference in their entirety.

TECHNICAL FIELD

[0002] The present disclosure generally to electrical electronic system, such as those including electrical connectors, used to interconnect electronic assemblies.

BACKGROUND

[0003] Electrical connectors are used in many electronic systems. It is generally easier and more cost effective to manufacture an electronic system as separate electronic assemblies, which may be joined together with electrical connectors. Electrical connectors may be used for interconnecting assemblies so that the assemblies may operate together as part of an electronic system. Electrical connectors, for example, may be mounted onto printed circuit boards within two assemblies that are connected by mating the electrical connectors. In other electronic systems, it may be impractical to join two printed circuit boards by directly mating electrical connectors on these printed circuit boards. For example, these printed circuit boards may be spaced so far apart in assembled electronic systems that the electrical connectors mounted in the boards cannot be directly connected.

[0004] In the electronic systems, assemblies may be interconnected through cables. The cables may be terminated with connectors that can mate with connectors mounted on printed circuit boards. In this way, the assemblies may be interconnected by plugging electrical connectors that are part of cable assemblies into electrical connectors mounted onto printed circuit boards. In other electronic system architectures, an electrical connector terminated to a cable may be mated with another electrical connector terminated to another cable.

[0005] Modern automobiles are examples of electronic systems with assemblies connected through cables. For example, the automobiles may include electronic control units (ECUs) for controlling various automotive systems, such as engines, transmission control units (TCUs), security systems, emission control, lighting, advanced driver assistance systems (ADASs), entertainment systems, navigation systems and cameras. The ECUs may be manufactured as separate assemblies that are connected by one or more cables routed between the assemblies. To simplify manufacturing, an assembly may include one or more cables terminated with an electrical connector, which can be mated to another electrical connector terminated to other cables or attached to a circuit board within another assembly.

SUMMARY

[0006] Aspects of the present disclosure relate to reliable high-speed, high-density connectors and interconnection systems thereof.

[0007] Some embodiments relate to an electrical connector. The electrical connector may include a housing assembly; a module held by the housing assembly and comprising a plurality of conductive pads along a mating edge disposed in a pitch less than or equal to 1.0 mm and a plurality of conductors extending away from the mating edge; and a connector position assurance (CPA) device coupled to the housing assembly and configured for securing the electrical connector to a mating electrical connector.

[0008] Optionally, the plurality of conductive pads are a plurality of first conductive pads configured for signal transmission; and the module comprises a second conductive pad configured for power transmission.

[0009] Optionally, the module may comprise a circuit board disposed inside the housing assembly and comprising the plurality of first conductive pads and the second conductive pad, each of the plurality of first conductive pads having a front end configured to contact a mating electrical connector and a rear end, the second conductive pad having a front end configured to contact the mating electrical connector and a rear end; a plurality of first cables comprising first wires disposed in a first pitch less than or equal to 1.0 mm, the first wires having ends mounted to the rear ends of the plurality of first conductive pads of the circuit board so as to electrically connect to the plurality of conductive pads; and a plurality of second cables comprising second wires disposed in a second pitch greater than the first pitch, the second wires comprising ends mounted to the rear end of the second conductive pad.

[0010] Optionally, the plurality of conductive pads conform to Peripheral Component Interconnect Express Generation 5 and/or Generation 6 for transmission rates.

[0011] Optionally, the CPA device comprises a first locking feature having a first locked position and a first unlocked position, and a second locking feature having a second locked position and a second unlocked position; the first locking feature and the second locking feature are disposed on different sides of the housing assembly; and the first locking feature is configured to be movable from the first locked position to the first unlocked position when the second locking feature is in the first unlocked position and to remain in the first locked position when the second locking feature is in the second locked position.

[0012] Optionally, the CPA device comprises a third locking features having a third locked position and a third unlocked position; and the third locking feature is coupled to the second locking feature such that the second locking feature is in the second locked position when the third locking feature is in the third locked position and the second locking feature is in the second unlocked position when the third locking feature is in the third unlocked position.

[0013] Optionally, the first locking feature comprises a spring biased element; and the second locking feature comprises a pull tab.

[0014] Some embodiments relate to an electrical connector. The electrical connector may include a housing assembly; a module held by the housing assembly and comprising a plurality of conductive pads; and a connector position assurance (CPA) device coupled to the housing assembly and configured for securing the electrical connector to a mating electrical connector, the CPA device comprising first and second locking features disposed on different sides of the housing assembly.

[0015] Optionally, the CPA device comprises a first locking feature having a first locked position and a first unlocked position, and a second locking feature having a second locked position and a second unlocked position; and the first locking feature is configured to be movable from the first locked position to the first unlocked position when the second locking feature is in the first unlocked position and to remain in the first locked position when the second locking feature is in the second locked position.

[0016] Optionally, the CPA device comprises a third locking features having a third locked position and a third unlocked position; and the third locking feature is coupled to the second locking feature such that the second locking feature is in the second locked position when the third locking feature

is in the third locked position and the second locking feature is in the second unlocked position when the third locking feature is in the third unlocked position.

[0017] Optionally, the CPA device comprises a first operating member configured to be operable in a mating direction so as to move the first locking feature between the first locked position and the first unlocked position; and a second operating member configured to be operable in a direction perpendicular to the mating direction so as to move the second locking feature between the second locked position and the second unlocked position.

[0018] Optionally, the third locking feature is configured to engage the first operating member when the third locking feature is in the third locked position so as to restrain movement of the first operating member, and disengage the first operating member when the third locked feature is in the third unlocked position.

[0019] Optionally, the first locking feature comprises a first locking head configured to engage a first side of the mating electrical connector; a first tail configured to engage the first operating member; and an intermediate portion joining the first locking head and the first tail and pivotably connected to the housing assembly about a first axis such that the first locking feature is pivotable between the first locked position and the first unlocked position.

[0020] Optionally, the first locking feature comprises a resilient beam configured to abut the housing assembly so as to bias the first locking head toward outside of the housing assembly; and the electrical connector comprises a resilient member disposed between the second locking feature and the housing assembly and configured to bias the second locking feature toward the second locked position.

[0021] Optionally, the second operating member is configured to engage a second side of the mating electrical connector; and the second operating member comprises the first locking feature, the second locking feature, and an intermediate portion joining the first locking feature and the second locking feature and pivotably connected to the housing assembly about a second axis perpendicular to the first axis.

[0022] Optionally, the plurality of conductive pads are dimensioned to be in electrical contact with the mating electrical connector as long as the first locking feature is in the first locked position.

[0023] Some embodiments relate to an electrical connector. The electrical connector may include a main housing comprising a mating portion for engaging a mating electrical connector; a plurality of conductive terminals held by the main housing; a cage disposed outside the main housing and comprising a front portion extending beyond the mating portion; and an outer housing attached to the front portion of the cage, the outer housing comprising first and second locking features configured to engage complementary locking features of the mating electrical connector, the first locking feature disposed on a first side of the outer housing and the second locking feature disposed on a second side of the outer housing.

[0024] Optionally, the cage comprises a barb protruding outwardly; and the outer housing comprises an opening for engaging the barb.

[0025] Optionally, the outer housing comprises a locking portion extending beyond the front portion of the cage, the locking portion comprising the first and second locking features; the locking portion of the outer housing comprises first walls opposite each other and a second wall joining the first walls, the second wall comprising a bulge protruding outwardly; the first locking feature comprises first openings extending through the first walls; and the second locking feature comprises a second opening extending through the bulge.

[0026] Optionally, the main housing comprises a slot configured to receive a substrate comprising a plurality of conductive pads disposed in a pitch less than or equal to 1.0 mm.

[0027] Some embodiments relate to an automotive wiring harness. The automotive wiring harness may include the electrical connector described herein.

[0028] Some embodiments relate to an electrical connector. The electrical connector may comprise a housing assembly, a circuit board, a plurality of high-speed data cables, and a connector position

assurance (CPA) device. The circuit board may be disposed inside the housing assembly, and include a plurality of conductive pads for electrical contact with a mating electrical connector. The plurality of high-speed data cables may have ends mounted onto the circuit board and electrically connected to the plurality of conductive pads. The CPA device may be disposed on the housing assembly, and configured for locking the electrical connector to the mating electrical connector. [0029] Optionally, the plurality of high-speed data cables may conform to Peripheral Component Interconnect Express Generation 5 and/or Generation 6 for transmission rates.

[0030] Optionally, a pitch between adjacent high-speed data cables of the plurality of high-speed data cables may be less than or equal to 1.0 mm.

[0031] Optionally, the CPA device may include a first locking feature and a second locking feature. The first locking feature may have a first locked position to lock the mating electrical connector and a first unlocked position to release the mating electrical connector. The second locking feature may have a second locked position to lock the mating electrical connector and a second unlocked position to release the mating electrical connector. The first locking feature and the second locking feature may be configured for locking different sides of the mating electrical connector, respectively.

[0032] Optionally, the CPA device may further include a first operating member and a third locking feature. The first operating member may be operable along a predetermined direction. The first locking feature may be movable from the first locked position to the first unlocked position based on the operation of the first operating member. The third locking feature may have a third locking position and a third unlocked position. The third locking feature may be configured to lock the first operating member when in the third locked position such that the first locking feature is retained in the first locked position; and the third locking feature may be further configured to release the first operating member when in the third unlocked position such that the first operating member is operable.

[0033] Optionally, the CPA device may include a first operating member, a first locking feature, and a third locking feature. The first operating member may be operable along a predetermined direction. The first locking feature may have a first locked position to lock the mating electrical connector and a first unlocked position to release the mating electrical connector. The first locking feature may be movable from the first locked position to the first unlocked position based on the operation of the first operating member. The third locking feature may have a third locking position and a third unlocked position. The third locking feature may be configured to lock the first operating member when in the third locked position such that the first locking feature is retained in the first locked position; and the third locking feature may be further configured to release the first operating member when in the third unlocked position such that the first operating member is operable.

[0034] Optionally, the CPA device may further include a second locking feature, having a second locked position to lock the mating electrical connector and a second unlocked position to release the mating electrical connector. The first locking feature and the second locking feature may be configured for locking different sides of the mating electrical connector, respectively.

[0035] Optionally, the first locking feature may include a first locking head for locking with the mating electrical connector, a first tail for coupling to the first operating member, and an intermediate portion joining the first locking head and the first tail. The intermediate portion may be pivotably connected to the housing assembly about a first predetermined axis such that the first locking feature is pivotable between the first locked position and the first unlocked position. When the first locking feature is moved from the first locked position to the first unlocked position, the first locking head may be pivoted toward the inside of the housing assembly.

[0036] Optionally, the first locking feature may include a resilient beam bent toward the inside of the housing assembly, and the resilient beam may abut against the housing assembly so that the first locking head is biased toward the outside of the housing assembly.

[0037] Optionally, the first operating member may be configured to be returnable to an initial position for locking with the third locking feature under the bias of the resilient beam.

[0038] Optionally, the first tail may have a V-shaped slot with an opening facing the inside of the housing assembly. The first operating member may be slidable along a sidewall of the V-shaped slot such that the first locking feature is pivotable between the first locked position and the first unlocked position.

[0039] Optionally, an end of the first tail may abut against the housing assembly when the first locking feature is in the first locked position.

[0040] Optionally, the CPA device may further include a second operating member. An intermediate portion of the second operating member may be pivotably connected to the housing assembly about a second predetermined axis, and the second locking feature and the third locking feature may be located at opposite ends of the second operating member, respectively. When the second locking feature is in the second locked position, the third locking feature may be in the third locked position; and when the second locking feature is in the second unlocked position, the third locking feature may be in the third unlocked position.

[0041] Optionally, the predetermined direction may be perpendicular to the second predetermined axis.

[0042] Optionally, when the second locking feature is in the second unlocked position, the electrical connector is able to be partially mated/unmated from the mating electrical connector. The electrical connector may be completely detached from the mating electrical connector based on the operation of the first operating member.

[0043] Optionally, the second locking feature may be retained in the second unlocked position under the action of the mating electrical connector during the partial mating/unmating of the electrical connector.

[0044] Optionally, the plurality of conductive pads of the circuit board may be dimensioned to be in electrical contact with the mating electrical connector during the partial mating/unmating of the electrical connector.

[0045] Optionally, the housing assembly may comprise a first housing and a second housing fixed onto the first housing, and the first locking feature and the first operating member may be mounted between the first housing and the second housing.

[0046] Optionally, the first locking feature may be pivotably connected between the first housing and the second housing through a first shaft. The first shaft may pass through the first locking feature, and two ends of the first shaft may be fixed to the first housing and the second housing, respectively, such that the first housing and the second housing are connected to each other.

[0047] Optionally, there may be a pair(s) of the first locking features. Each pair of the first locking features may be disposed on two sides of the first housing and the second housing opposite each other along a first lateral direction perpendicular to the predetermined direction. The first locking features each may be mounted between the first housing and the second housing through a respective first shaft.

[0048] Optionally, the first housing may include a first mating portion and a first cable connection portion. The first mating portion may be configured for mating with the mating electrical connector and include a first mounting channel for accommodating the circuit board. The second housing may be fastened onto the first cable connection portion to form a second mounting channel with the first cable connection portion. The plurality of high-speed data cables may pass through the second mounting channel. The first operating member may be slidably connected to the second mounting channel along the predetermined direction by the first housing and the second housing.

[0049] Optionally, the first mating portion and the first cable connection portion may be disposed opposite each other along the predetermined direction.

[0050] Optionally, the housing assembly may further comprise a third housing fastened onto the second housing. The second operating member may be mounted between the second housing and

the third housing. The second locking feature may pass through the second housing and protrudes beyond an outer surface of the second housing. The third locking feature may be located inside the third housing.

[0051] Optionally, second shafts may be provided on two sides of the intermediate portion of the second operating member, respectively. Holes may be encircled by the second housing and the third housing. The second shafts may be inserted into the holes, respectively.

[0052] Optionally, the second operating member may include a second operating portion passing through the second housing and located between the second shaft and the second locking feature.

[0053] Optionally, the housing assembly may further comprise a strain relief member, connected to the second housing and spaced apart from the first shaft along the predetermined direction. A portion of the first housing may be clamped between the second housing and the strain relief member.

[0054] Optionally, the second housing may be embedded with a nut. A fastener may pass through the strain relief member and be threaded to the nut to secure the strain relief member to the second housing.

[0055] Optionally, a resilient member may be provided between the second operating member and the housing assembly, for biasing the second locking feature toward the second locked position.

[0056] Optionally, the housing assembly may include a first slot and a second slot extending along the predetermined direction, and a first operating portion. The first slot and the second slot may be spaced apart along a first lateral direction perpendicular to the predetermined direction. The first operating member may include a first arm and a second arm, movably inserted into the first slot and the second slot, respectively. One end of the first arm and that of the second arm may be coupled to the first locking feature such that the first locking feature is movable between the first locked position and the first unlocked position. The first operating portion may be connected between the other end of the first arm and that of the second arm, and protrude beyond the housing assembly.

[0057] Optionally, the first operating member may further include a crossbeam connected between the middle portions of first arm and the second arm, and the third locking feature may be hooked onto the crossbeam when in the third locked position.

[0058] Optionally, the housing assembly may include an opening, and the crossbeam may pass through the opening to limit a sliding range of the first operating member.

[0059] Optionally, the predetermined direction may be a mating direction of the electrical connector to the mating electrical connector.

[0060] Some embodiments relate to a connector position assurance (CPA) device. The CPA device may comprise a first locking feature and a second locking feature. The first locking feature may have a first locked position to lock a mating electrical connector and a first unlocked position to release the mating electrical connector. The second locking feature may have a second locked position to lock the mating electrical connector and a second unlocked position to release the mating electrical connector. The first locking feature and the second locking feature may be configured for locking different sides of the mating electrical connector, respectively.

[0061] Optionally, the CPA device may further include a first operating member and a third locking feature. The first operating member may be operable along a predetermined direction. The first locking feature may be movable from the first locked position to the first unlocked position based on the operation of the first operating member. The third locking feature may have a third locking position and a third unlocked position. The third locking feature may be configured to lock the first operating member when in the third locked position such that the first locking feature is retained in the first locked position; and the third locking feature may be further configured to release the first operating member when in the third unlocked position such that the first operating member is operable.

[0062] Some embodiments relate to a connector position assurance (CPA) device. The CPA device

may comprise a first operating member, a first locking feature and a third locking feature. The first operating member may be operable along a predetermined direction. The first locking feature may have a first locked position to lock a mating electrical connector and a first unlocked position to release the mating electrical connector. The first locking feature may be movable from the first locked position to the first unlocked position based on the operation of the first operating member. The third locking feature may have a third locking position and a third unlocked position. The third locking feature may be configured to lock the first operating member when in the third locked position such that the first locking feature is retained in the first locked position; and the third locking feature may be further configured to release the first operating member when in the third unlocked position such that the first operating member is operable.

[0063] Optionally, the CPA device may further include a second locking feature, having a second locked position to lock the mating electrical connector and a second unlocked position to release the mating electrical connector. The first locking feature and the second locking feature may be configured for locking different sides of the mating electrical connector, respectively.

[0064] Optionally, the first locking feature may include a first locking head for locking with the mating electrical connector, a first tail for coupling to the first operating member, and an intermediate portion joining the first locking head and the first tail. The intermediate portion may be pivotably connected to the housing assembly about a first predetermined axis such that the first locking feature is pivotable between the first locked position and the first unlocked position.

[0065] Optionally, the first locking feature may include a bent resilient beam configured to bias the first locking feature toward the first locked position.

[0066] Optionally, the first operating member may be configured to be returnable to an initial position for locking with the third locking feature under the bias of the resilient beam.

[0067] Optionally, the first tail may have a V-shaped slot. The first operating member may be slidable along a sidewall of the V-shaped slot such that the first locking feature is pivotable between the first locked position and the first unlocked position.

[0068] Some embodiments relate to an electrical connector. The electrical connector may comprise a main housing, a plurality of conductive terminals, a cage, and an outer housing. The main housing may include a second mating portion for mating with a mating electrical connector and a second mounting portion. The plurality of conductive terminals may be held by the main housing and extend from the second mating portion to the second mounting portion. The cage may surround outside the main housing, and have a front portion extending beyond the second mating portion. The outer housing may be attached to the front portion of the cage and include a complementary locking feature configured to engage with a locking feature of the mating electrical connector.

[0069] Optionally, the cage may be formed by bending a metal sheet. Edges of the metal sheet opposite each other may be formed with a mortise and tenon structure such that the edges of the metal sheet are interconnected with each other.

[0070] Optionally, the cage may include a barb protruding outwardly, and the outer housing may be provided with an opening for engaging with the barb.

[0071] Optionally, the cage may include a board lock located outside of the outer housing.

[0072] Optionally, an inner surface of the cage and an outer surface of the second mating portion may be spaced apart to form a gap for receiving a first mating portion of the mating electrical connector.

[0073] Optionally, the outer housing may be sleeved over the front portion of the cage, and the outer housing may include a locking portion extending beyond the front end and have the complementary locking feature disposed thereon.

[0074] Optionally, the second mating portion and the second mounting portion may be perpendicular to each other.

[0075] Optionally, the outer housing may have a first dimension along a first lateral direction and a second dimension along a second lateral direction. The first dimension may be greater than the

second dimension. The first lateral direction and the second lateral direction may be perpendicular to each other and perpendicular to a mating direction of the electrical connector to the mating electrical connector. The outer housing may have first walls opposite each other along the first lateral direction and second walls opposite each other along the second lateral direction. The second walls may have a bulge outwardly protruding. The complementary locking feature may include first openings in the first walls and a second opening in the bulge.

[0076] The techniques described herein may be used alone or in any suitable combination. The foregoing summary is provided by way of illustration and is not intended to be limiting.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0077] The accompanying drawings are not intended to be drawn to scale. In the drawings, each identical or nearly identical component that is illustrated in various figures may be represented by a like numeral. For purposes of clarity, not every component may be labeled in every drawing. In the drawings:

[0078] FIG. 1A is a perspective view of an electronic system comprising a first electrical connector and a mating second electrical connector connected fully mated with each other, according to some embodiments.

[0079] FIG. 1B is a perspective view of the electronic system of FIG. 1A, illustrating the first electrical connector and the mating second electrical connector partially mated/unmated with each other.

[0080] FIG. 1C is a side view of the electronic system of FIG. 1B.

[0081] FIG. 1D is a top view of the electronic system of FIG. 1B.

[0082] FIG. 1E is a perspective view of the electronic system of FIG. 1A, illustrating the first electrical connector and the mating second electrical connector separated from each other.

[0083] FIG. 1F is a partially enlarged cross-sectional view of the electronic system of FIG. 1A.

[0084] FIG. 1G is a partially enlarged cross-sectional view of the electronic system of FIG. 1B.

[0085] FIG. 1H is a partially enlarged cross-sectional view of the electronic system of FIG. 1E.

[0086] FIG. 2A is a top, front perspective view of the first electrical connector of the electronic system of FIG. 1A.

[0087] FIG. 2B is a partially enlarged view of the first electrical connector of FIG. 2A, illustrating a second operating member without a pressing force applied thereon.

[0088] FIG. 2C is a partially enlarged view of the first electrical connector of FIG. 2B, illustrating the second operating member with a pressing force applied thereon.

[0089] FIG. 2D is a top view of the first electrical connector of FIG. 2A.

[0090] FIG. 2E is a partially enlarged cross-sectional view of the electrical connector of FIG. 2B.

[0091] FIG. 2F is a partially enlarged cross-sectional view of the first electrical connector of FIG. 2C.

[0092] FIG. 2G is a partially exploded view of the first electrical connector of FIG. 2A.

[0093] FIG. 2H is a top, rear perspective view of the first electrical connector of FIG. 2A, with a portion thereof exploded.

[0094] FIG. 2I is a bottom, rear perspective view of the first electrical connector of FIG. 2A, with another portion thereof exploded.

[0095] FIG. 3 is a perspective view of the first electrical connector of FIG. 2A, illustrating a third housing and the second operating member hidden.

[0096] FIG. 4A is a perspective view of a module of the first electrical connector of FIG. 2A.

[0097] FIG. 4B is a perspective view of the module of FIG. 4A, with a module housing hidden.

[0098] FIGS. 5A-5C are perspective, side and top views of a first locking feature of the first

electrical connector of FIG. 2A, respectively.

[0099] FIGS. 6A-6C are perspective, side and top views of a second operating member of the first electrical connector of FIG. 2A, respectively.

[0100] FIG. 7 is a perspective view of a first housing of the first electrical connector of FIG. 2A.

[0101] FIG. 8 is a perspective view of the first electrical connector of FIG. 2A, with a first operating member exploded.

[0102] FIG. 9 is a perspective view of the first electrical connector of FIG. 2A, with the first operating member assembled.

[0103] FIGS. 10A-10B are simplified views of the first housing of FIG. 7 with respect to the first operating member.

[0104] FIG. 11A is a top, rear perspective view of a second electrical connector of the electronic system of FIG. 1A.

[0105] FIG. 11B shows a cross-sectional perspective view of the second electrical connector of FIG. 11A.

[0106] FIG. 12A is a perspective view of a cage of the second electrical connector of FIG. 11A.

[0107] FIG. 12B is a bottom, rear perspective view of the cage of FIG. 12A.

[0108] FIG. 13A is an exploded perspective view of the second electrical connector of FIG. 11A, with portions hidden, showing a main housing and conductive terminals.

[0109] FIG. 13B is a perspective view of the second electrical connector of FIG. 13A.

[0110] FIG. 14 is a perspective view of an outer housing of the second electrical connector of FIG. 11A.

[0111] The above accompanying drawings include the following reference signs:

[0112] **100**, first electrical connector; **110**, housing assembly; **111**, first housing; **110A**, first mating portion; **110B**, first body; **1111**, first mounting channel; **1112**, first cable connection portion; **1113**, first slot; **1114**, second slot; **112**, second housing; **1121**, opening; **1122**, mounting opening; **1123**, aperture; **1124**, snap slot; **1125**, first cut; **113**, third housing; **1131**, snap; **1131A**, flat surface; **1131B**, guide bevel; **1132**, second cut; **114**, strain relief member; **1141**, cross member; **1142**, break; **115**, second positioning post; **120**, circuit board; **121A**, first end; **121B**, second end; **122**, conductive pad; **122A**, first conductive pad; **122B**, second conductive pad; **123**, inclined surface; **124**, slot; **131**, high-speed data cable; **131A**, end of high-speed data cable; **132**, power cable; **141**, first locking feature; **1411**, first locking head; **1411A**, hook; **1411B**, guide; **1411C**, outer surface; **1411E**, opening; **1412**, first tail; **1412A**, first sidewall; **1412B**, second sidewall; **1413**, first intermediate portion; **1413A**, hole; **1414**, resilient beam; **142**, second locking feature; **143**, third locking feature; **144**, first operating member; **1441**, first operating portion; **1442**, opening; **1443**, pillar; **1444**, crossbeam; **1445**, first arm; **1446**, second arm; **145**, second operating member; **1451**, pressing portion; **1452**, second shaft; **1454**, first positioning post; **150**, first shaft; **160**, resilient member; **170**, module housing; **200**, second electrical connector; **210**, main housing; **211**, second mating portion; **211A**, groove; **211B**, second mating end; **211C**, second mounting end; **212**, second mounting portion; **2121**, protrusion; **2122**, latch; **220**, conductive terminal; **221**, mating end; **222**, mounting end; **223**, intermediate portion; **223A**, curved portion; **230**, cage; **231**, mortise and tenon; **232**, barb; **233**, board lock; **233A**, first protrusion; **233B**, second protrusion; **234**, front portion; **235**, resilient sheet; **240**, outer housing; **2401**, first wall; **2402**, second wall; **2402A**, bulge; **241**, locking portion; **242**, opening; **251**, first complementary locking feature; **252**, second complementary locking feature; **260**, gap.

DETAILED DESCRIPTION

[0113] The Inventors have recognized and appreciated connector design techniques that enable high-speed, high-density connectors and interconnection systems thereof that may operate reliably in the harsh environment presented by an automobile. Techniques described herein may provide hybrid electrical connectors that can be used to maintain signal integrity for high-speed signal transmission in harsh environments, such as those encountered in vehicles, while providing reliable

power supply. Techniques described herein may enable economically assembling such hybrid electrical connectors.

[0114] According to aspects of the present disclosure, an interconnection system may include an electrical connector and a mating electrical connector. The electrical connector may include a housing assembly, a module, and a connector position assurance (CPA) device. The module may include densely disposed conductive pads. The module may include a circuit board with the conductive pads and high-speed data cables having ends mounted onto the conductive pads. The CPA device may be disposed on the housing assembly, and configured for securing the electrical connector to the mating connector. The CPA device may include locking features disposed on different sides of the housing assembly, which can improve reliability. The mating electrical connector may include an outer housing having complementary locking features. The connectors can be electrically connected as long as a first locking feature in its locked position. Such techniques can enable interconnection systems for harsh environments such as one presented by an automobile.

[0115] In some embodiments, an electrical connector may comprise a first locking feature and a second locking feature, which may be used to position the electrical connector to a mating electrical connector on different sides when the two are mated with each other. The first locking feature and the second locking feature may provide position assurance along different directions, so that a reliable connection can be provided with respect to vibrations in various directions.

[0116] In some embodiments, the first locking feature and the second locking feature may be provided with a first operating member and a second operating member, respectively, for unlocking. The first operating member and the second operating member may have different operation manners and/or different operation directions. The locking features may be unlocked by different manners, so that the probability that the electrical connectors are disconnected due to accidental external factors can be reduced. Exemplary manners include pushing in, pressing, pulling, rotating, etc. Optionally, the first operating member and the second operating member may be operated in different manners. Optionally, the first operating member and the second operating member may be operated in different directions. In some embodiments, a combination of operation manners may make the electrical connectors compact in structure and achieve economical assembly of the electrical connectors, e.g., a combination of pressing and an outward pulling.

[0117] In some embodiments, a third locking feature may be provided for one or more of the operating members. The third locking feature may be configured to avoid mis-operation, which may lead to accidental unlocking of the corresponding locking feature. In some embodiments, the third locking feature may be provided for the pulling-type first operating member. It may be troublesome for a user to operate simultaneously two or more members, and particularly in a narrow or cable-intensive space. In some embodiments, the second locking feature and the third locking feature may be designed to be linked, such that when the second operating member is operated to unlock the second locking feature, the third locking feature can also be unlocked, and thus the first operating member can be released so as to be in an operable state. Optionally, the electrical connector may be configured such that when the second locking feature is unlocked, the electrical connector can move to a partially mated/unmated position with respect to the mating electrical connector. The third locking feature may also be maintained in its unlocked position, and the user may freely operate the first operating member to unlock the first locking feature, realizing one-handed unlocking operation.

[0118] In some embodiments, a mating electrical connector may include locking features that are complementary to the aforementioned first locking feature and the second locking feature. The mating electrical connector may include a main housing that holds a plurality of conductive terminals, an outer housing, and a cage connecting the main housing and the outer housing. In some embodiments, a front portion of the cage may extend beyond the main housing. The outer housing may be connected to the front portion of the cage and include a locking portion extending

beyond the front portion of the cage. The locking portion of the outer housing may include the locking features. In some embodiments, the locking portion of the outer housing may include first walls opposite each other and a second wall joining the first walls. The second wall may include a bulge protruding outwardly. The first walls may include first openings extending therethrough as locking features. The bulge of the second wall may include a second opening extending therethrough as a locking feature.

[0119] FIGS. 1A to 1E illustrate a portion of an electronic system, for example, used in a vehicle, for interconnecting a plurality of electronic devices in the electronic system. As shown, the electronic system may comprise a first electrical connector **100** and a second electrical connector **200** that are detachably mated to each other. The second electrical connector **200** may be mounted onto a first circuit board (not shown). The first electrical connector **100** may be a cable connector. The first electrical connector **100** may be electrically connected via cables to an electronic device (not shown), such as a second circuit board. The first electrical connector **100** and the second electrical connector **200** may provide an interconnection between the electronic device and the first circuit board. The electronic device may be spaced apart from the first circuit board by a distance. For example, the second electrical connector **200** is a socket electrical connector mounted onto the first circuit board. The first electrical connector **100**, as a plug electrical connector, may be mated with the socket electrical connector. In some embodiments, the first electrical connector may be mounted directly onto a circuit board, and the second electrical connector is configured as a cable connector, so as to interconnect a further electronic device with the circuit board. Optionally, both the first electrical connector and the second electrical connector are connected to respective electronic devices, such as circuit boards, via cables. One of the first electrical connector and the second electrical connector may be fixed to a target electronic device. In harsh environments such as those presented by an automobile, the electronic system may provide high-speed data transmission while withstanding vibrations.

[0120] FIGS. 2A-2I illustrate the first electrical connector **100** according to some embodiments of the present disclosure. For clear and concise description, a mating direction X-X, a first lateral direction Y-Y and a second lateral direction Z-Z are shown in the drawings. The mating direction X-X may be the direction in which the first electrical connector **100** is mated to the second electrical connector **200**. The first lateral direction Y-Y and the second lateral direction Z-Z may be both perpendicular to the mating direction X-X, and the first lateral direction Y-Y and the second lateral direction Z-Z may be perpendicular to each other.

[0121] In the illustrated example, the first electrical connector **100** may include a housing assembly **110**. The housing assembly **110** may include a first mating portion **110A**. The first mating portion **110A** is configured to match the second electrical connector **200** in shape, such that the first electrical connector **100** can be precisely positioned onto the second electrical connector **200** once they are mated together. The conductors in the first electrical connector **100** and the second electrical connector **200** can be reliably and correspondingly electrically connected. In addition, the housing assembly **110** may further comprise a first body **110B**. The housing assembly **110** may be molded with an insulating material, such as plastic. Plastic may include but not limited to liquid crystal polymer (LCP), polyphenylene sulfite (PPS), high-temperature nylon or poly-p-phenylene oxide (PPO), or polypropylene (PP), or other materials may be used. In some cases, plastic may be thermosetting plastic. In some cases, insulating plastic may include such as fiberglass-reinforced insulating material.

[0122] The first electrical connector **100** may include a circuit board **120**, as shown in FIGS. 4A and 4B. The circuit board **120** may be disposed inside the housing assembly **110**, and the circuit board **120** includes a plurality of conductive pads **122** for electrical contact with a mating electrical connector. A substrate of the circuit board **120** may be made of a fiberglass-reinforced material. Optionally, the substrate may be made of Teflon, in the case that the circuit board **120** is used for transmitting high-speed data signals. The conductive pads **122** may be formed on the circuit board

120 by printing or machining. Adjacent conductive pads **122** are spaced suitably to reduce the risk of short circuit and crosstalk. Optionally, metal layers may be formed on the surfaces of the conductive pads **122** to avoid poor contact due to oxidation. In some embodiments, the conductive pads **122** may be gold fingers arranged densely. Optionally, the circuit board **120** may be substantially accommodated inside the first mating portion **110A** of the housing assembly **110**. When the first electrical connector **100** are mated to the second electrical connector **200**, the circuit board **120** may be inserted into the second electrical connector **200** such that the conductive pads **122** on the circuit board **120** are in electrical contact with corresponding conductive elements of the second electrical connector **200**. The density of conductive pads **122** on the circuit board **120** can be increased as needed, so as to satisfying the need for high-density signal transmission. The circuit board **120** may include a first end **121A** in the interior of the housing assembly **110** and a second end **121B** facing the second electrical connector **200** along the mating direction X-X. The second end **121B** may be chamfered so as to form an inclined surface **123** that serves as a guide during insertion into the mating electrical connector.

[0123] The plurality of conductive pads **122** in the first electrical connector **100** may be used to transmit high-speed data signals and/or power signals. Optionally, the conductive pads **122** may be used entirely for transmitting high-speed data signals or entirely for transmitting power signals. Optionally, a portion of the conductive pads **122** may be used for transmitting high-speed data signals and the others for transmitting power signals. Conductive pads **122** may transmit electrical power when transmitting power signals.

[0124] The first electrical connector **100** may further include a plurality of high-speed data cables **131**. Ends **131A** of the plurality of high-speed data cables **131** may be attached to the circuit board **120** and in electrical contact with the plurality of conductive pads **122**. In the illustrated example, at least a portion of the conductive pads **122** are used to transmit high-speed data signals. The ends **131A** of the high-speed data cables **131** may be connected to pads on the first end **121A** of the circuit board **120** by, for example, brazing, spot welding or ultrasonic welding, such that the ends **131A** are reliably fixed and electrically connected to the pads. The pads are connected with respective conductive pads **122** via conductive traces in the circuit board **120** such that the plurality of high-speed data cables **131** are electrically connected to the plurality of conductive pads **122**. The plurality of high-speed data cables **131** may extend through the first body **110B** of the housing assembly **110** to the first mating portion **110A**. In the illustrated example, the high-speed data cables **131** extend substantially along the mating direction X-X. Optionally, the high-speed data cables may be bent, for example, to 90 degrees or any other suitable angle. In this case, the first body **110B** may have a curved portion to be adapted to bends of the high-speed data cables. The ends **131A** of the high-speed data cables **131** together with the first end **121A** of the circuit board **120** may be cladded in the housing assembly **110**, so as to avoid short circuit caused by moisture, oxidization or foreign matters, and to avoid the high-speed data cables **131** from separation from the circuit board **120** by external forces.

[0125] The high-speed data cables **131** may generally have electrical properties superior to low-speed data cables for transmitting low-speed data signals. Accordingly, the first electrical connector **100** may be backwardly compatible for transmitting low-speed data signals, such as low frequency signals (e.g., with a frequency of less than 66 MHz), signals with lower data rates (e.g., less than 100 Mb/s). The low-speed data signals may typically include auxiliary information and/or control signals responsible for supporting and managing data transmission, rather than primary data. The control signals may include one or more of control commands, status information, error detection and correction, and the like. The auxiliary information may include one or more of checksums, flow control, clock synchronization, and the like. The auxiliary information may improve data integrity, reliability and synchronization. Optionally, the first electrical connector **100** may be used to transmit high-speed data signals or low-speed data signals. Optionally, the first electrical connector **100** may be used to transmit both high-speed data signals and low-speed data signals.

[0126] The first electrical connector **100** may include a power cable **132**. As shown in FIGS. **4A** and **4B**, the conductive pads **122** on the circuit board **120** may include a plurality of first conductive pads **122A** spaced apart mutually and electrically connected to the high-speed data cables **131**, so as to cooperate with the mating electrical connector to provide transmission of high-speed data. The conductive pads **122** may include a wide second conductive pad **122B** that may be electrically connected to the power cable **132** and cooperate with the mating electrical connector to transmit electrical power. In some embodiments, the power cable **132** may include a plurality of thin cables in parallel rather than a single thick cable, such that the difficulties of assembling and soldering can be reduced. For example, each cable has a conductive portion with a circular cross-section, and narrower than a portion of the second conductive pad **122B** connected to the power cable **132**. A thick cable may need to be flattened before connected to the second conductive pad **122B** to enlarge contact area and lower contact resistances. This results in a more complex process and difficulty in ensuring quality. The thick cable also causes an increased dimension of the first electrical connector **100** in the second lateral direction Z-Z. In comparison, where the power cable **132** includes a plurality of thin cables connected to the second conductive pad **122B** in parallel, the dimension may be increased in the first lateral direction Y-Y only. Since the first electrical connector **100** may have space to accommodate the power cable **132** in the first lateral direction Y-Y, the outline dimensions of the first electrical connector **100** may be unchanged. The plurality of thin cables may also ensure the reliability of the power supply. Even if one or more of the cables are disabled, the power supply may not be interrupted completely. Optionally, there may be a plurality of second conductive pads spaced apart from one another. In some embodiments, the circuit board **120** may include a slot **124**, which divides the circuit board **120** into a power portion and a high-speed data portion. The second conductive pad **122B** may be disposed on a front side and/or a reverse side of the power portion and the first conductive pads **122A** may be disposed on a front side and/or a reverse side of the high-speed data portion. As the first electrical connector **100** is mated with the second electrical connector **200**, the slot **124** may fit to a projection of the mating electrical connector so as to be dummy-proof. The slot **124** may also separate the second conductive pad **122B** for transmitting electrical power, from the first conductive pads **122A** for transmitting high-speed data to some extent, thereby minimizing interference. In some embodiments, the slot **124** may be formed by two spaced apart circuit boards.

[0127] The first electrical connector **100** may include a connector position assurance (CPA) device, which is disposed on the housing assembly **110**. In some embodiments, the CPA device may adopt one of latch, threaded connection and snap, or any combination of the above, provided that the first electrical connector **100** can be locked to the second electrical connector **200** when they are mated. The first electrical connector **100** can be protected from potential accidentally detaching from the mating electrical connector due to vibrations or external forces.

[0128] The first electrical connector **100** may include the circuit board **120**. The circuit board **120** may include the conductive pads **122** for electrical contact with the mating second electrical connector. In contrast to each contact being fabricated as a separate conductor mounted into the housing assembly **110**, the density of conductive pads **122** can be increased by fabricating the conductive pads **122** on the circuit board **120**, but the dimensions of the first electrical connector **100** and the mating second electrical connector **200** are not significantly altered. Moreover, the conductive pads **122** can be precisely positioned on the circuit board **120** and different batches of the products can have higher consistency. Signal integrity is improved even when high-speed data signals are transmitted. Signal transmission may not be affected even in a harsh environment (e.g., such as one presented by an automobile), since the first electrical connector **100** is in reliable electrical connection to the mating second electrical connector **200** by the CPA device.

[0129] In some embodiments, the plurality of high-speed data cables **131** may conform to Peripheral Component Interconnect Express Generation 5 (PCIe Gen 5) and/or Generation 6 (PCIe Gen 6) for transmission rates. High-speed data transmission usually requires high frequency signals

as carrier waves to ensure sufficient bandwidth. High frequency signals may be significantly affected by many factors, such as conductor impedance, skin effect. Moreover, external interference may also seriously affect the high-speed data transmission, resulting in data loss. In this regard, the high-speed data cables **131** can adopt cables with shield layers connected to the signal grounds and are in pairs for transmitting differential signals, to eliminate signal interference from external electric and magnetic fields to a certain extent. Meanwhile, the electric and magnetic fields of the cables in each pair may mutually offset, and reduced electromagnetic radiation is less likely to interfere with other devices. The high-speed data cables **131** may be simplified in the drawings, without showing the structural details, such as shield layers.

[0130] In some embodiments, a pitch between adjacent high-speed data cables **131** is less than or equal to 1 mm to meet the requirements of high-density cables, which provide a prerequisite for high-density signal transmission. The first electrical connector **100** may support high bandwidth data transmission and be backwardly compatible for low bandwidth data transmission. For example, for devices that communicate at high bandwidths, all high-speed data cables **131** may be used. For devices that communicate at low bandwidths, a portion of the high-speed data cables **131** may be used, with the rest being idle. In some applications, in some embodiments, a portion of the high-speed data cables **131** in a device A may be used to receive data from a device B and another portion of the high-speed data cables **131** may be used to transmit data to the device B, thereby receiving and transmitting data simultaneously, and realizing full-duplex communication.

[0131] Referring back to FIG. 1A, in some embodiments, the CPA device may include a first locking feature **141**. The first locking feature **141** has a first locked position for locking the second electrical connector **200** and a first unlocked position for releasing the second electrical connector **200**. In some embodiments, the CPA device may also include a second locking feature **142**. The second locking feature **142** has a second locked position for locking the second electrical connector **200** and a second unlocked position for releasing the second electrical connector **200**. The first locking feature **141** and the second locking feature **142** may be applied on different sides of the second electrical connector **200**, respectively. High stresses may be generated between the locking features of the first electrical connector **100** and complementary locking features of the mating electrical connector when subjected to external forces. If the locking features are disposed on the same side of the first electrical connector **100**, the deterioration of the CPA device and the complementary locking features on the mating electrical connector may be accelerated in a harsh environment, such as one presented by an automobile, resulting in the failure of the CPA device. The vibrations may be unidirectional and regular in such a harsh environment. Thus, once the vibration direction substantially coincides with the locking direction of the locking features, the locking features may be failed collectively. In addition, in a trend of miniaturization, the first electrical connector **100** is not adequate for accommodating both the first locking feature **141** and the second locking feature **142** on the same side, and the same is true for the second electrical connector **200**. The mechanical strength of the first electrical connector **100** and the second electrical connector **200** can be less impacted by disposing the first locking feature **141** and the second locking feature **142** on different sides of the first electrical connector **100**. As shown in FIGS. 2A and 2B, the first electrical connector **100** may have a larger dimension in the first lateral direction Y-Y than that in the second lateral direction Z-Z. The first locking feature **141** may be disposed on sides of the first electrical connector **100** opposite in the first lateral direction Y-Y. Correspondingly, the second locking feature **142** may be disposed on one or two of sides of the first electrical connector **100** opposite in the second lateral direction Z-Z. As a result, three-point locking is possible. In this case, the second locking feature **142** may be dimensionally larger than the first locking feature **141**. When the second locking feature **142** is disposed on one side of the first electrical connector **100**, the second locking feature **142** may include a larger locking projection that can withstand a pulling force which is at least greater than that a single first locking feature **141** can withstand. The larger second locking feature **142** can withstand a larger pulling

force and evenly spread that pulling force to avoid stress concentration that may lead to deformation and fracture. Optionally, the second locking feature **142** may be disposed on both sides of the first electrical connector **100** opposite in the second lateral direction Z-Z. The first electrical connector **100** may become relatively complex in structure, and the dimension may become greater in the second lateral direction Z-Z. These locking features may be redesigned in number and arrangement by those skilled in the art, as appropriate.

[0132] In some embodiments, the first locking feature **141** and the second locking feature **142** may be movable between their respective locked positions and unlocked positions. In order to differentiate, the locked positions of the first locking feature **141** and the second locking feature **142** may be referred to as a first locked position and a second locked position, respectively, and their unlocked positions may be referred to as a first unlocked position and a second unlocked position, respectively. When the first electrical connector **100** is connected in place with the second electrical connector **200** as shown in FIG. 1A, the first locking feature **141** and the second locking feature **142** may be coupled to the first complementary locking feature **251** and the second complementary locking feature **252** on the second electrical connector **200**, respectively. In some embodiments, the first complementary locking feature **251** and the second complementary locking feature **252** may be configured as openings, grooves, and/or stop projections, provided that they are capable of coupling to the first locking feature **141** and the second locking feature **142**, respectively. When the first electrical connector **100** vibrates in an operating environment, the CPA device of the first electrical connector **100** may substantially withstand component forces of the external forces in the mating direction X-X, and component forces in directions perpendicular to the mating direction X-X may be borne substantially by the second electrical connector **200** and the first mating portion **110A** of the housing assembly **110** inserted into the second electrical connector **200**. In some embodiments, the first locking feature **141** and the second locking feature **142** may withstand the component forces in the mating direction X-X in different periods. For example, when one of the first locking feature **141** and the second locking feature **142** fails, the other can still work. There may be one or more first locking features **141** and second locking features **142**. Optionally, when the first electrical connector **100** is subjected to a pulling force, the second locking feature **142** is first subjected to the component forces in the mating direction X-X. When the second locking feature **142** is in the second locked position, the second locking feature **142** may be engaged with the second complementary locking feature **252**, such as notch or opening, to reduce the risk of relative displacement of the first electrical connector **100** to the second electrical connector **200**. When the second locking feature **142** is in the second unlocked position, the second locking feature **142** is disengaged from the second complementary locking feature **252** on the second electrical connector **200**. In this case, the first electrical connector **100** may be movable relative to the second electrical connector **200** in the mating direction X-X within a certain range, and the first locking feature **141** may still couple to the corresponding first complementary locking feature **251** on the second electrical connector **200**. When the first electrical connector **100** is moved to a boundary of the above range, the first electrical connector **100** is in a partially mated/unmated position, as shown in FIGS. 1B to 1D. At this point, although the second locking feature **142** is disengaged from the second complementary locking feature **252**, the first locking feature **141** is engaged with the first complementary locking feature **251**, such that the first electrical connector **100** is still connected with the second electrical connector **200**. In some embodiments, the first electrical connector **100** and the second electrical connector **200** may remain in electrical contact when the first electrical connector **100** is moved within the range along the mating direction X-X. If the first locking feature **141** is in the first unlocked position and the external forces are continuously applied on the first electrical connector **100** in the mating direction X-X, the first electrical connector **100** may be completely detached from the second electrical connector **200**. FIG. 1E illustrates the first electrical connector **100** in a complete detachment state. [0133] FIGS. 1F to 1H show partial enlarged views of the first electrical connector **100** in mated

position, partially mated/unmated position, and complete detachment position, respectively, with respect to the second electrical connector **200**. In some embodiments, the first locking feature **141** may include a hook **1411A** protruding along the first lateral direction Y-Y beyond the housing assembly **110**. When the first locking feature **141** is in the first locked position and the second locking feature **142** is in the second locked position, as shown in FIG. 1F, the hook **1411A** may extend into the first complementary locking feature **251**, such as opening. When the second locking feature **142** is in the second unlocked position, the first locking feature **141** may be limited by the first complementary locking feature **251** such that the first electrical connector **100** is movable between the mated position with the second electrical connector **200**, as shown in FIG. 1F, and the partially mated/unmated position, as shown in FIG. 1G. Referring to FIG. 1G, in the partially mated/unmated position, the hook **1411A** is hooked at the edge of the first complementary locking feature **251**, so that even if the second locking feature **142** fails during the operation, the first electrical connector **100** and the second electrical connector **200** can still be kept in a connected state by the first locking feature **141**. When the first electrical connector **100** need to be completely detached from the second electrical connector **200**, the first locking feature **141** can be placed in the first unlocked position, as shown in FIG. 1H, such that the hook **1411A** can be disengaged from the first complementary locking feature **251** of the second electrical connector **200**. Thereupon, the first electrical connector **100** can be completely detached from the second electrical connector **200**, as shown in FIG. 1E. In some embodiments, the first locking feature **141** may be unlocked based on the operation of the first operating member as described below.

[0134] In some embodiments, the CPA device may include a first operating member **144** operable along a predetermined direction. The first locking feature **141** may be moved from the first locked position to the first unlocked position by operating the first operating member **144**. The first operating member **144** may have any other suitable structure, such as button, disposed on the housing assembly **110**. In the illustrated example, the first operating member **144** may include a pull tab. As shown in FIG. 1E, the first operating member **144** may be mounted onto the first body **110B** of the housing assembly **110**. The pull tab may be disposed on a side of the first body **110B** opposite the first mating portion **110A**. In some embodiments, the predetermined direction may be the mating direction X-X of the first electrical connector **100** with the mating second electrical connector **200**. In this case, the first locking feature **141** may be unlocked by pulling the first operating member **144** outwardly along the mating direction X-X, such that the first electrical connector **100** can be detached from the second electrical connector **200** whereby. This is more in line with operating habits of a user. Optionally, the first operating member may also be operated by rotating, pressing, etc., in which case the predetermined direction may be at an angle to the mating direction X-X of the first electrical connector **100**, or even opposite the pull-out direction of the first electrical connector **100**. In some embodiments, the first operating member **144** may be coupled to the first locking feature **141**. As the user operates the first operating member **144**, the first locking feature **141** may be moved from the first locked position to the first unlocked position by a mechanical structure, such as bulge, oblique surface, and gear. In a natural state, the first locking feature **141** may be retained in the first locked position, which can reduce the risk of the first locking feature **141** failing in the partially mated/unmated state and cause the first electrical connector **100** to completely detach from the second electrical connector **200** accidentally. The number of parts can be reduced and the structure can be simplified by this mechanical linkage design, thereby to reduce dimensions and ensure reliable operation.

[0135] FIGS. 2A to 2I illustrate a first electrical connector **100** according to an exemplary embodiment of the present disclosure. Optionally, a second locking feature **142** of the first electrical connector **100** may be a snap. Referring to FIG. 1B and FIG. 2C in conjunction, the second locking feature **142** is moved toward the inside of the first electrical connector **100** under resilience, as the first electrical connector **100** is inserted into the second electrical connector **200**. The second locking feature **142** may automatically reset and be engaged with the second

complementary locking feature **252** from the inside of the second electrical connector **200** when the first electrical connector **100** is connected in place to the second electrical connector **200**, referring to FIGS. **1A**, **2A**, and **2C**.

[0136] Optionally, the first electrical connector **100** may include a third locking feature **143**, as shown in FIGS. **2E** and **2F**. The third locking feature **143** serves to lock the first operating member **144**, in particular to lock the first operating member **144** when the first locking feature **141** is in the first locked position. The third locking feature **143** may have a third locked position and a third unlocked position. The third locking feature **143** is configured to lock the first operating member **144** when in the third locked position, such that the first locking feature **141** is retained in the first locked position, as shown in FIG. **2E**. The third locking feature **143** is also configured to release the first operating member **144** when in the third unlocked position, enabling the first operating member **144** to be operable, as shown in FIG. **2F**, such that the first operating member **144** may be pulled toward the right to unlock the first locking feature **141**. Although the first operating member **144** is linked with the first locking feature **141** in the illustrated example, the first operating member **144** can be locked by the third locking feature **143**. Accordingly, the first locking feature **141** may not be accidentally unlocked by undesired movement of the first operating member **144** under external forces caused by vibrations, such that the first electrical connector **100** may be reliably attached to the mating second electrical connector **200**. Also, the first operating member **144** may be always in motion in the harsh environment, if the first operating member **144** is not locked in a desired position by the third locking feature **143**. In the illustrated example, the first operating member **144** may not move frequently unless it is operated, thus prolonging its service life.

[0137] Optionally, the CPA device may include the first locking feature and the third locking feature. For example, the first electrical connector **100** may be locked to the mating electrical connector by the first locking feature. When the first locking feature is in the first unlocked position, the electrical connector **100** can be completely detached from the mating electrical connector. The first operating member can be protected from potential abnormal movement by the third locking feature, and the first locking feature can, in turn, be retained at the locked position, thereby enabling the first electrical connector **100** to be reliably connected to the mating electrical connector.

[0138] Optionally, the CPA device may include the first locking feature and the second locking feature, for locking to different sides of the second electrical connector. For example, the CPA device may not include the third locking feature. The first operating member may be retained in an initial position by a resilient member or any other suitable approach. The first locking feature may be actuated to the first unlocked position, when subjected to greater external forces along the predetermined direction. This may be achieved by replacing the third locking feature with a structure, such as spring. Comparatively, the embodiments including the third locking feature may be more reliable.

[0139] In some embodiments, as shown in FIGS. **1A**, **1B**, **2A** and **2G**, the first locking feature **141** may include a first locking head **1411** for locking with the mating second electrical connector **200**, a first tail **1412** for coupling to the first operating member **144**, and a first intermediate portion **1413** joining the first locking head **1411** and the first tail **1412**. The first intermediate portion **1413** is pivotably connected to the housing assembly **110** about a first predetermined axis P-P (refer to FIG. **5C**), such that the first locking feature **141** is pivotable between the first locked position and the first unlocked position. In some embodiments, as shown in FIGS. **5A-5B**, the first intermediate portion **1413** may be provided with holes **1413A**. The first predetermined axis P-P is defined by the holes **1413A** and a shaft fixed to the housing assembly **110**. Optionally, the shaft may be fixed to the first intermediate portion **1413** and the holes may be disposed in the housing assembly **110**. When the first locking feature **141** is moved from the first locked position to the first unlocked position, the first locking head **1411** is pivoted toward the interior of the housing assembly **110**. In

some embodiments, the first locking head **1411** may include a protrusion, such as the hook **1411A**, as shown in FIGS. **1F-1G** and FIGS. **5A-5C**. The first locking head **1411** may have an outer surface **1411C** toward the outside of the housing assembly **110**. The hook **1411A** may protrude from the outer surface **1411C** of the first locking head **1411**. As the first electrical connector **100** are connected to the second electrical connector **200** in place, the first locking feature **141** is in the first locked position, and the hook **1411A** may protrude into the first complementary locking feature **251** of the second electrical connector **200**, as shown in FIG. **1F**. The outer surface **1411C** of the first locking head **1411** may abut against or be spaced apart from an inner surface of the second electrical connector **200**. When the first electrical connector **100** is moved to the partially mated/unmated position with respect to the second electrical connector **200** along the mating direction X-X, as shown in FIG. **1G**, the hook **1411A** is engaged with the edge of the first complementary locking feature **251** of the second electrical connector **200**, such that the first electrical connector **100** can discontinue to move along the mating direction X-X with respect to the second electrical connector **200** and stop in the partially mated/unmated position. As the first locking feature **141** is moved to the first unlocked position under the action of the first operating member **144**, the first locking head **1411** is pivoted toward the interior of the housing assembly **110**, and thereupon the hook **1411A** can be freed from the restriction of the first complementary locking feature **251**. At this point, the first electrical connector **100** is completely detached from the second electrical connector **200** by moving the first electrical connector **100** along the mating direction X-X continuously.

[0140] In some embodiments, the first locking feature **141** may be made of sheet metal. The first locking head **1411** may be elastically deformable such that the first locking head **1411** may be pivoted inwardly when the first electrical connector **100** is inserted into the second electrical connector **200**. The first locking head **1411** may restore to its natural state when entering the first complementary locking feature **251**.

[0141] In some embodiments, the first locking head **1411** may include a resilient beam **1414** bent toward the interior of the housing assembly **110**. The resilient beam **1414** may abut against the housing assembly **110** so that the first locking head **1411** is biased toward the exterior of the housing assembly **110**. In this way, the first locking feature **141** can be retained in the first locked position. In some embodiments, an opening **1411E** may extend from the first intermediate portion **1413** to the first locking head **1411**. The resilient beam **1414** may extend from an edge of the opening **1411E** far away from the first locking head **1411**. Continuously referring to FIGS. **5A-5C**, the hook **1411A** described above may also be disposed at another edge of the opening **1411E**. In some embodiments, the hook **1411A** and the resilient beam **1414** may be disposed on two opposite edges of the opening **1411E**, respectively. Optionally, the opening **1411E** may also be seated in the first locking head **1411**. This may shorten the length of the resilient beam **1414**. Optionally, there may be more than one resilient beam. As shown in FIG. **5B**, the resilient beam **1414** is disposed on a side of the first predetermined axis P-P closer to the first locking head **1411**. The resilient beam **1414** can apply a torque onto the first locking head **1411** around the first predetermined axis P-P toward the exterior of the housing assembly **110**, as shown in FIGS. **1F** to **1H**, such that the first locking head **1411** is biased toward the exterior of the housing assembly **110** in its natural state and retained in the first locked position. A free end of the resilient beam **1414** may be bent away from the housing assembly **110** such that a smooth surface of the bent portion contacts the housing assembly **110**. In this way, the contact surface can be increased, and the contact surface is relatively smooth, such that defects such as burrs may not be introduced by processing. The surface of the housing assembly **110** is uneasily to be scratched during the movement of the resilient beam **1414** relative to the housing assembly **110** along the mating direction X-X, thereby effectively prolonging the service life.

[0142] Optionally, the first operating member **144** may be configured to return to its initial position under the bias of the resilient beam **1414**. The first operating member **144** may be locked with the

third locking feature **143** at the initial position. In this way, when the first electrical connector **100** is completely detached from the second electrical connector, the first operating member **144** can automatically return to the initial position. The first operating member **144** may be designed to structurally fit with the housing assembly **110**, such that a gap therebetween may be reduced when the first operating member **144** is in the initial position and foreign objects, such as the risk of dusts entering into the first electrical connector **100** to affect the service life may be reduced. Optionally, the first operating member **144** may also return to the initial position under external forces exerted by the user.

[0143] Optionally, the first locking head **1411** may have a guide **1411B** at its front end. The guide **1411B** may be bent in a direction opposed to the protruding direction of the hook portion **1411A**. During the first electrical connector **100** mating with the second electrical connector **200**, the guide **1411B** may facilitate the entry of the first locking head **1411** into the second electrical connector, and the first locking feature **141** may be pivoted toward the interior of the housing assembly **110** under the guide **1411B**, to ensure a smooth insertion.

[0144] In some embodiments, the first tail **1412** may have a V-shaped slot, and an opening of the V-shaped slot toward the interior of the housing assembly **110**, as shown in FIGS. 1F to 1H. The first operating member **144** slides along a sidewall of the V-shaped slot to allow the first locking feature **141** to be pivoted between the first locked position and the first unlocked position. The V-shaped slot has a first sidewall **1412A** and a second sidewall **1412B**. The first sidewall **1412A** is connected between the first intermediate portion **1413** and the second sidewall **1412B**. The first operating member **144** is slidable along the second sidewall **1412B**. When the first locking feature **141** is in the first locked position, a portion of the first operating member **144** that slides along the second sidewall **1412B** may be located just at the deepest part of the V-shaped slot. The first tail **1412** of the first locking feature **141** may tend to be pivoted about the first predetermined axis P-P toward the interior of the housing assembly **110** under the resilient beam **1414**. A distal end of the first tail **1412** may abut against the housing assembly **110** or against the first operating member **144**, and thereupon the resilient beam **1414** may exert a certain pressure on the housing assembly **110** or the first operating member **144** via the distal end of the first tail **1412**, such that the first locking feature **141** is retained in the first locked position. By operating the first operating member **144**, for example, moving the first operating member **144** along an upward direction in the drawings, a pressure is exerted to the second sidewall **1412B** of the V-shaped slot, so that the first locking feature **141** can overcome the resilience of the resilient beam **1414** and be pivoted from the first locked position to the first unlocked position around the first predetermined axis P-P. When the first operating member **144** is released, the first locking feature **141** is pivoted to the first locked position under the resilience of the resilient beam **1414**, and the second sidewall **1412B** of the V-shaped slot can drive the first operating member **144** back to the initial position.

[0145] In some embodiments, the distal end of the first tail **1412** may abut against the housing assembly **110**, rather than the first operating member **141**, when the first locking feature **141** is in the first locked position. It should be appreciated that if the distal end of the first tail **1412** abuts against the first operating member **144**, the distal end of the first tail **1412** may scrape the surface of the first operating member **144** when the first operating member **144** is actuated relative to the first tail **1412** along the mating direction X-X, such that the first operating member **144** may be worn severely where the distal end of the first tail **1412** is rough. In comparison, the first tail **1412** would not move relative to the housing assembly **110** along the mating direction X-X. It is easier to provide a first operating member **144** with an opening **1442**, as shown in FIG. 8, through which the first tail **1412** passes to contact with the housing assembly **110**. In this way, the first operating member **144** is avoided from scratching, and the costs can be substantially reduced. Moreover, a pillar **1443** formed between the opening **1442** and the end of the first operating member **144** is just able to serve as the portion of the first operating member **144** that slides on the second sidewall **1412B**. The pillar **1443** will be discussed in more detail hereinafter.

[0146] In some embodiments, the CPA device may also include a second operating member **145**, as shown in FIGS. 2E and 2F, and FIGS. 6A to 6C. An intermediate portion of the second operating member **145** is pivotably connected to the housing assembly **110** about a second predetermined axis Q-Q, and the second locking feature **142** and the third locking feature **143** are located at opposite ends of the second operating member **145**, respectively. In some embodiments, the second operating member **145** may include a pressing portion **1451**. The pressing portion **1451** may pass through the housing assembly **110** to be externally operable. The pressing portion **1451** may be disposed between the second predetermined axis Q-Q and the second locking feature **142**. In some embodiments, the second operating member **145** may be a member integrally molded with the second locking feature **142** and the third locking feature **143**, or manufactured separately and then joined with the second locking feature **142** and the third locking feature **143**. The pressing portion **1451** may protrude beyond the surface of the housing assembly **110** of the first electrical connector **100** for easy operation. Referring back to FIGS. 2B and 2C, the second locking feature **142** may protrude beyond the outer surface of the housing assembly **110** when in the second locked position, so that the second locking feature **142** can be engaged with the second electrical connector **200**. When the pressing portion **1451** may be pressed, as shown in FIG. 2C, the second locking feature **142** is moved into the housing assembly **110** to reach the second unlocked position. In some embodiments, as shown in FIGS. 6A to 6D, the second operating member **145** may be provided with second shafts **1452** concentric with the second predetermined axis Q-Q, such that the second operating member **145** is pivotably connected to the housing assembly **110** via the second shafts **1452**.

[0147] The pressing portion **1451** is pivotable in the same direction as the first locking feature **141** and in an opposite direction to the third locking feature **143**. When the second locking feature **142** is in the second locked position, the third locking feature **143** is in the third locked position, as shown in FIG. 2E. When the pressing portion **1451** is pressed, the second locking feature **142** is pivoted downwardly to the second unlocked position and disengaged from the second electrical connector, such that the first electrical connector **100** is unlocked by the second locking feature **142**. At the same time, the third locking feature **143** is pivoted upwardly to reach the third unlocked position. For example, when the second locking feature **142** is in the second locked position, the third locking feature **143** is in the third locked position; and when the second locking feature **142** is in the second unlocked position, the third locking feature **143** is in the third unlocked position. For example, the second locking feature **142** and the third locking feature **143** may synchronously move between respective locked and unlocked positions. In this way, the second locking feature **142** and the third locking feature **143** can be unlocked by use of the pressing portion **1451** on the second operating member **145**, without an additional unlocking operating member.

[0148] In some embodiments, a resilient member **160** may be provided between the second operating member **145** and the housing assembly **110**, for biasing the second locking feature **142** toward the second locked position. It should be appreciated that, when the pressing portion **1451** does not protrude beyond the surface of the housing assembly **110**, it may be difficult to manually restore the pressing portion **1451** to its original position after it is pressed. In order to reliably lock the first electrical connector **100** to the mating electrical connector, the second locking feature **142** may be biased toward the second locked position in a natural state. The second locking feature may be maintained in the second locked position by the resilient member **160** and the pressing portion **1451** is protruding for easy operation. The resilient member **160** may include any suitable part, such as resilient sheet, spring. In some embodiments, a magnetic force may be applied onto the second locking feature **142** for biasing. In some embodiments, the resilient member **160** may be disposed corresponding to the second locking feature **142**. Thus, the resilient force generated by the resilient member **160** may act directly on the second locking feature **142**, ensuring that the second locking feature **142** is reliably in the second locked position. In some embodiments, the second operating member **145** may be provided with a first positioning post **1454**, the housing assembly

110 may be provided with a second positioning post **115**, and two ends of the resilient member **160** may be sleeved onto the first positioning post **1454** and the second positioning post **115**, respectively.

[0149] In some embodiments, the predetermined direction along which the first operating member **144** is operable may be perpendicular to the second predetermined axis Q-Q. In the illustrated example, the second predetermined axis Q-Q may be parallel to the first lateral direction Y-Y, and the predetermined direction may be parallel to the mating direction X-X. The housing assembly **110** may have sufficient space in the first lateral direction Y-Y to accommodate the second operating member **145** pivotable about the second predetermined axis Q-Q, and it is more advantageous for the third locking feature **143** of the second operating member **145** to lock the movement perpendicular to the second predetermined axis Q-Q. It should be appreciated that, once the first operating member **141** is applied a force in the predetermined direction parallel to the mating direction X-X, and the force may be conducted to the second operating member **145** via the third locking feature **143**. The force on the second operating member **145** is perpendicular to the second shafts **1452** thereof, and therefore the second shafts **1452** can spread the force to the housing assembly **110** without causing the second operating member **145** to be twisted. If the predetermined direction is parallel to the second predetermined axis Q-Q, the force applied onto the first operating member **144** and in turn onto the second operating member **145** in operation may be not perpendicular to the second shafts **1452**, which may result in deformations of the second shafts **1452** and/or a portion of the housing assembly **110** for accommodating the second shafts **1452**. Accordingly, the second operating member **145** may be loose. In addition, when the second locking feature **142** on the second operating member **145** is locked to the mating electrical connector and the predetermined direction is parallel to the mating direction X-X, if a pulling force is applied onto the first operating member **144**, the second shafts **1452** can withstand a part of the pulling force conducted to the second operating member **145**, and the second locking feature **142** can withstand another part of the pulling force, allowing for more reliable locking.

[0150] In some embodiments, the second locking feature **142** may be retained in the second unlocked position by the mating electrical connector during the partial mating/unmating of the first electrical connector **100**. In some narrower environments, the user is able to detach the first electrical connector **100** with one hand. In these applications, the user is typically complete one action at a time. If the second operating member **145** automatically returns to its original position after being operated, causing the second locking feature **142** and the third locking feature **143** to be re-locked after being unlocked, the user cannot continue to operate the first operating member **144** to detach the first electrical connector **100** from the mating second electrical connector **200**. As shown in FIG. 1B, the user may press the second operating member **145** as well as pull out the first electrical connector **100** along the mating direction X-X, so as to move the first electrical connector **100** to the partially mated/unmated position. At this point, the second locking feature **142** is blocked by the mating electrical connector and cannot restore to the second locked position. Therefore, the user may then operate the first operating member **144** to completely detach the first electrical connector **100** from the second electrical connector **200**, and no longer operate the second operating member **145**. Thus, the operating difficulty can be obviously reduced based on the above technical solution, so that the first electrical connector **100** can be applied to a variety of complex and narrow environments.

[0151] In some embodiments, the plurality of conductive pads **122** on the circuit board **120** may be dimensioned to maintain electrical contact with the mating electrical connector during the partial mating/unmating of the first electrical connector **100**. As described above, it is undesirable that the first electrical connector **100** and the mating electrical connector are accidentally disconnected by external forces. Even if the second locking feature **142** fails due to external factors, such as bumps and collisions, which cause the first electrical connector **100** to reach the partially mated/unmated position, at least the most basic function can be ensured. In some embodiments, the lengths of the

circuit board **120** and the conductive pads **122** thereon may compensate for a distance between the mated position and the partially mated/unmated position, such that at least a part of conductors of the electrical connector **100** can remain electrically connected to the mating electrical connector. Optionally, the length of some of the conductive pads on the circuit board is slightly shorter than that of the others, and the shorter conductive pads can be disconnected electrically from the mating electrical connector during the partial mating/unmating, while the other longer conductive pads are retained electrically connected. The longer conductive pads may be connected to circuits for dedicatedly detecting the disconnection of the electrical connectors, and/or be multiplexed by a part of conductive pads for the ground cables for shielding, and/or be connected to some data cables. Power and/or signals between the devices employing the electrical connectors may not be completely interrupted at the partially mated/unmated stage of the electrical connectors. The devices may detect the partial mating/unmating of the electrical connectors, and thus issue an alarm or perform other operations.

[0152] In some embodiments, the housing assembly **110** may include a first housing **111** and a second housing **112** fastened onto the first housing **111**, as shown in FIGS. 2A, 2G, and 2H. The first locking feature **141** and the first operating member **144** may be mounted between the first housing **111** and the second housing **112**. The first housing **111** and the second housing **112** may be processed by injection molding, respectively. In this way, molds for the first housing **111** and the second housing **112** can become simpler in structure, production and design costs can be reduced, and yields may be improved. Moreover, the difficulty of assembling the first locking feature **141** and the first operating member **144** can be reduced.

[0153] In some embodiments, the first locking feature **141** is pivotably connected between the first housing **111** and the second housing **112** through a first shaft **150**. The first shaft **150** passes through the first locking feature **141**. Two ends of the first shaft **150** are inserted to the first housing **111** and the second housing **112**, respectively. As described above, the first locking feature **141** may include the holes **1413A**. The first shaft **150** may pass through the holes **1413A** (in FIG. 5) and the diameter of the holes **1413A** may be slightly larger than that of the first shaft **150**, such that the first locking feature **141** may be pivoted around the first shaft **150** with less resistance. In some embodiments, the first shaft **150** may also fit closely with the holes **1413A**, and the first shaft **150** and the first locking feature **141** may be pivoted together. In some embodiments, the first shaft **150** may be made of a material with plasticity, such as an aluminum tube. When the first shaft **150** is not required to be pivoted together with the first locking feature **141**, the ends of the first shaft **150** may be fixed to the first housing **111** and the second housing **112** by interference fit or fasteners, etc. The ends of the first shaft **150** may be squeezed and expanded so as to increase their diameters, such that the ends cannot be divorced from the apertures of the first housing **111** and the second housing **112**. The first housing **111** and the second housing **112** may be fixed to each other.

[0154] In some embodiments, there may be a pair(s) of the first locking features **141** disposed on opposite sides of the first housing **111** and the second housing **112** in the first lateral direction Y-Y perpendicular to the predetermined direction. The first locking features **141** each is mounted between the first housing **111** and the second housing **112** through the respective first shaft **150**. The first electrical connector **100** may be mated with the second electrical connector **200** with one pair of the first locking features **141**. When the first electrical connector **100** is subjected to external forces along the first lateral direction Y-Y, at least one of the first locking features **141** can withstand the external forces. The first electrical connector **100** may not tilt which may cause misalignment of the conductive pads **122**. Moreover, the pair of the first locking features **141** may also share the external forces applied on the end of the first housing **111** inserted into the second electrical connector **200**, and the end of the first housing **111** may not break. One of the first locking features **141** is shown in FIGS. 5A to 5C, and the other one is a mirror image of the first locking features **141** shown in the drawings. This makes the design less difficult and the external forces are evenly dispersed. However, the present disclosure also does not exclude embodiments in

which the first locking features have different structures and are not mirror images of each other. [0155] In some embodiments, the aforementioned first mating portion **110A** for mating with the mating electrical connector may be formed by the first housing **111**, and the first housing **111** may include a first cable connection portion **1112**. In some embodiments, the shape of the first mating portion **110A** may be tubular, and the cross-section thereof may be a circle, an oval, or a rounded rectangle, etc. An outer surface of the first mating portion **110A** may be smooth, without sharp edges or sharp corners, making it safer for the user. The first mating portion **110A** may match a mating portion (e.g., a second mating portion described below) of the mating electrical connector in shape, such that the first electrical connector **100** can be precisely positioned relative to the mating electrical connector even subjected to external forces or vibrations. The first mating portion **110A** may include a first mounting channel **1111**, as shown in FIGS. 2A, 2G and 8. The circuit board **120** may be fixed inside the first mounting channel **1111**. The second housing **112** is fastened onto the first cable connection portion **1112** and forms a second mounting channel with the first cable connection portion **1112**. The plurality of high-speed data cables **131** pass through the second mounting channel. The first operating member **144** is limited by the first housing **111** and the second housing **112** to be slidably connected to the second mounting channel along the predetermined direction. As shown in FIG. 3, an upper surface of the first operating member **144** may abut against an inner wall of the second housing **112**, and a lower surface of the first operating member **144** may abut against an inner wall of the first housing **111**, thereby positioning the first operating member **144** along the second lateral direction Z-Z. The circuit board **120** may be fixed inside the first mounting channel **1111** by a module housing **170**, as shown in FIGS. 2G and 4A. The module housing **170** may be molded on the circuit board **120** and may have a certain degree of resilience. When the first electrical connector **100** is connected to the mating electrical connector, the module housing **170** may undergo an appropriate deformation, to reduce the risk of the circuit board **120** being damaged if the circuit board **120** is forcibly inserted into the mating electrical connector.

[0156] In some embodiments, the first mating portion **110A** and the first cable connection portion **1112** may be opposite along the predetermined direction, which may be parallel to the mating direction X-X. When the user expects to unlock the first electrical connector **100**, the first operating member **144** is moved relative to the housing assembly **110** along the predetermined direction, and the first electrical connector **100** can be detached from the mating electrical connector by continuing to move the first operating member **144**. It can be simple and convenient to achieve with one hand. The first mating portion **110A** and the first cable connection portion **1112** are opposite along a straight line, and thus there is no need to bend the cables. Accordingly, the structure is simpler and production and assembly costs are lower.

[0157] In some embodiments, as shown in FIGS. 2G and 8, the first housing **111** may include a first slot **1113** and a second slot **1114** extending along the predetermined direction (e.g., the mating direction X-X) and spaced apart along the first lateral direction Y-Y. The first operating member **144** may include a first arm **1445** and a second arm **1446** slidably inserted into the first slot **1113** and the second slot **1114**, respectively. As shown in FIGS. 8 and 9, the first arm **1445** and the second arm **1446** of the first operating member **144** may extend along the mating direction X-X. The first slot **1113** and the second slot **1114** may be integrally molded in the first housing **111**. The spacing between the first slot **1113** and the second slot **1114** matches with the spacing between the first arm **1445** and the second arm **1446**, and the width (e.g., the dimension in the first lateral direction Y-Y) of the first slot **1113** may be slightly larger than the width of the first arm **1445** such that a fit-up gap may be reserved between the two, the dimension of the fit-up gap may be 0.1-0.5 mm. Similarly, the width of the second slot **1114** may be slightly larger than the width of the second arm **1446**, so that the second arm **1446** can slide easily inside the second slot **1114** and not swing in the second slot **1114**. In some embodiments, to facilitate mounting of the first arm **1445** and the second arm **1446**, the first slot **1113** and the second slot **1114** may extend to the surface of the first

housing **111** facing the second housing **112**. After the second housing **112** is fastened to the first housing **111**, the first arm **1445** and the second arm **1446** may be limited so that they can slide along the first slot **1113** and the second slot **1114**, respectively. In some embodiments, the first arm **1445** and the second arm **1446** can be kept in respective positions, in the case that the first housing **111** has a greater rigidity. The widths of the first slot **1113** and the second slot **1114** may not match the widths of the first arm **1445** and the second arm **1446**, respectively. As shown in the example illustrated in FIG. **10A**, outer side surfaces of the first arm **1445** and the second arm **1446** opposed in the first lateral direction Y-Y may respectively abut against two sidewalls of the first slot **1113** and the second slot **1114** that are proximate to the outer surface of the first housing **111**, whereas inner side surfaces of the first arm **1445** and the second arm **1446** facing to each other may respectively have gaps from the sidewalls of the first slot **1113** and the second slot **1114**. The first operating member **144** may not swing inside the first housing **111**. FIG. **10B** illustrates another example in which the inner side surfaces of the first arm **1445** and the second arm **1446** facing to each other along the first lateral direction Y-Y may respectively abut against the two sidewalls of the first slot **1113** and the second slot **1114** that are proximate to the inside of the first housing **111**, whereas the outer side surfaces of the first arm **1445** and the second arm **1446** may be spaced apart from the sidewalls of the first slot **1113** and the second slot **1114**. This also enables the first operating member **144** to be stable. The above two examples may require less precision in the machining of the first slot **1113** and the first arm **1445**, as well as the second slot **1114** and the second arm **1446**, so as to reduce costs.

[0158] The spacing between the first arm **1445** and the second arm **1446** may be larger, such that the first arm **1445** and the second arm **1446** may be positioned at the locations near the outer surface of the first housing **111**. When the first operating member **144** is subjected to external forces in the first lateral direction Y-Y, the larger spacing may serve to spread the forces more evenly across the first housing **111**, reducing the risk of deformation of the first operating member **144** and the first housing **111**. In some embodiments, the sidewalls of the first slot **1113** and the second slot **1114** should have sufficient thickness, thereby ensuring the structural strength of the first slot **1113** and the second slot **1114**. On this basis, the spacing between the first arm **1445** and the second arm **1446** may be as large as possible, so as to respectively match with the first arm **1445** and the second arm **1446** which are spaced apart at a larger distance. Moreover, the space between the first arm **1445** and the second arm **1446** may also allow the high-speed data cables **131** and the power cables **132** to be mentioned below to pass through, and/or be used to house the third locking feature **143** and the second operating member **145**, etc.

[0159] Optionally, the first slot **1113** and the second slot **1114** may also be formed in the second housing **112**, or formed by both the first housing **111** and the second housing **112**.

[0160] Ends (referred to as coupling ends) of the first arm **1445** and the second arm **1446** toward the first mating portion **110A** are coupled to the first locking features **141**, such that the first locking features **141** are movable between the respective first locked position and first unlocked position. The coupling ends of the first arm **1445** and the second arm **1446** may protrude beyond the first slot **1113** and the second slot **1114**, respectively, to couple to the first locking features **141** on both sides, respectively.

[0161] The first operating member **144** may also include a first operating portion **1441**, connected between ends of the first arm **1445** and the second arm **1446** opposite the first mating portion **110A**. The first operating portion **1441** extends beyond the housing assembly **110**. In some embodiments, the first operating portion **1441** may be a handle that can be operated or pulled by a user. The first operating portion **1441** may be a pull tab with an aperture for receiving the user's finger or a tool. The user can operate the first operating member **144** along the predetermined direction by passing the finger or tool through the aperture. Compared with embodiments in which the first operating member is moved by friction from fingers, the first operating portion **1441**, e.g., a pull tab, may be less prone to slippage, and the user can easily operate the first operating member

1441 even if there are snags. The first operating portion **1441** may be integrally processed with the first operating member **144** by the same material in molding process. The first operating portion **1441** may also be made by a different material from the rest of the first operating member **144**. In this case, the first operating portion **1441** may be made of resilient materials, such as rubber, and mounted onto the first operating member **144** by a suitable method such as gluing, fusing, fastener. The first operating portion **1441** may be over-molded onto the first operating member **144**. The first operating portion **1441** in a resilient material can effectively improve the user experience during operation.

[0162] The user may pull the first operating portion **1441** outwardly along the predetermined direction, e.g., the mating direction X-X. The first operating portion **1441** may translate along the predetermined direction under the guidance of the first slot **1113** and the second slot **1114**. With reference to FIGS. **1F** to **1H** and FIGS. **8** to **9** in combination, a coupling end of the first arm **1445** may include the pillar **1443** mated with the V-shaped slot of the first tail **1412**. The pillar **1443** operates the first locking feature **141** by sliding along the second sidewall **1412B** of the V-shaped slot. The surface of the pillar **1443** contacting with the second sidewall **1412B** may be curved to minimize frictional resistance. The first locking feature **141** returns to the first locked position under the resilient beam **1414** when the first operating member **144** is released, and the inclined surface of the V-shaped slot may apply pressure to the pillar **1443**, causing the first operating member **144** to return to the initial position.

[0163] In some embodiments, the first operating member **144** may include a crossbeam **1444**, which is connected between middle portions of the first arm **1445** and the second arm **1446**, as in FIG. **2G** and FIGS. **8** to **9**. The third locking feature **143** may hook onto the crossbeam **1444** when in the third locked position, referring to FIGS. **2E** and **2F**. As described above, the first locking feature **141** may be unlocked by operating the first operating member **144**. In some embodiments, the third locking feature **143** may be in the shape of a hook. In some embodiments, the third locking feature **143** may be hooked at the middle of the crossbeam **1444** to uniformly apply a force to position the first operating member **144**, which in turn maintains the first locking feature **141** in the first locked position. Additionally, the mechanical strength of the first operating member **144** may also be enhanced by the crossbeam **1444** connected between the first arm **1445** and the second arm **1446**.

[0164] In some embodiments, the housing assembly **110** may include an opening **1121**, for example, formed in the inner surface of the second housing **112**, as shown in FIG. **3**. When the second housing **112** is mounted onto the first housing **111**, the crossbeam **1444** may pass through the opening **1121** to limit the sliding range of the first operating member **144**. As described above, the second sidewall **1412B** of the V-shaped slot of the first locking feature **141** can apply pressure to the pillar **1443** of the first operating member **144** under the force from the resilient beam **1414** when the first operating member **144** is released by the user, so that the first operating member **144** returns to its initial position. The risk of the pillar **1443** escaping from the V-shaped slot can be reduced and the first operating member **144** may not be moved too far to cause damages to the first locking feature **141** by limiting the first operating member **144** in the sliding range.

[0165] In some embodiments, the housing assembly **110** may comprise a third housing **113**, which may be fastened onto the second housing **112**. The second operating member **145** is mounted between the second housing **112** and the third housing **113**. The second housing **112** may be provided with an aperture **1123**, as shown in FIGS. **2G** and **2H**, and the second locking feature **142** passes through the aperture **1123** and protrudes beyond the outer surface of the second housing **112**. The third locking feature **143** is disposed on the inner side of the third housing **113**. As shown in FIG. **2H**, the third housing **113** may be provided with a snap **1131**. The third housing **113** may be mounted to the second housing **112** along the second lateral direction Z-Z. The snap **1131** may be resiliently deformed and be engaged with a recess of the second housing **112**. The second housing **112** may be provided with a snap slot **1124** to engage with the snap **1131**. For example, the snap

1131 may include a flat surface **1131A** parallel to the first lateral direction, and a guide bevel **1131B**. The guide bevel **1131B** may allow the third housing **113** to easily enter into the snap slot **1124** of the second housing **112**, while the flat surface **1131A** is engaged with the snap slot **1124**, such that the third housing **113** is fastened to the second housing **112**. After the assembly of the first housing **111** and the second housing **112**, the third locking feature **143** may be placed into a desired position through a mounting opening **1122** of the second housing **112**. Then, the mounting opening **1122** of the second housing **112** is closed by the third housing **113**, such that the risk of foreign objects falling into the mounting opening **1122** of the second housing **112** to cause failure of the third locking feature **143** may be reduced. Optionally, the third locking feature **143** may also be limited in the desired position by the third housing **113**.

[0166] In some embodiments, as described above, the second shafts **1452** may be provided on two sides of the intermediate portion of the second operating member **145**, respectively. In some embodiments, a rod may be inserted into holes in the second operating member **145** and the ends of the rod protrude from the side surfaces of the second operating member **145** to form the second shafts **1452**. The second shafts **1452** may be integrally molded with the second operating member **145**. Holes are encircled by the second housing **112** and the third housing **113** together, and the second shafts **1452** are inserted into the holes, respectively. In some embodiments, referring to FIGS. 2G and 3, the second housing **112** may include first cuts **1125**, and the third housing **113** may include second cuts **1132**. The openings of the first cuts **1125** and the second cuts **1132** may face to each other, such that they may form the holes. As described above, the third locking feature **143** may be mounted into the second housing **112** through the mounting opening **1122**. In the case that the third locking feature **143** is provided on the second operating member **145**, the second operating member **145** may be mounted in place through the mounting opening **1122**. The holes are formed for limiting the second shafts **1452** after the third housing **113** is mounted to the second housing **112**, such that the second operating member **145** is assembled easily. In this way, the second operating member **145** is pivotable around the second shafts **1452**.

[0167] In some embodiments, the housing assembly **110** may include a strain relief member **114**, as shown in FIGS. 2G, 2H, and 2I, the strain relief member **114** may be connected to the second housing **112** and spaced apart from the first shaft **150** along the predetermined direction. A portion of the first housing **111** is clamped between the second housing **112** and the strain relief member **114**. As described above, the first housing **111** and the second housing **112** may be fixed to each other by the first shaft **150**. When the first shaft **150** is close to one end of the first housing **111** and/or that of the second housing **112**, when their other ends are subject to external forces, the first shaft **150** may withstand a larger strain which may result in damages to one or more of the first housing **111**, the second housing **112**, and the first shaft **150**. The strain relief member **114** may be fixed to the end of the second housing **112** away from the first shaft **150** by any suitable means, such as fasteners, snap, to relieve the strain. As shown in FIG. 21, the strain relief member **114** may include a cross member **1141** which may apply a force in a second lateral direction Z-Z to a portion of the first housing **111**, such that the first housing **111** is pressed against the second housing **112**. The strain relief member **114** may include a break **1142**. As the strain relief member **114** is mounted to the first housing **111** and the second housing, the break **1142** forms a channel together with the inner wall of the second housing **112**. High-speed data cables **131** and/or power cables **132** may pass through the channel. The high-speed data cables **131** are more tensile on their own than where the high-speed data cables **131** connect to the circuit board **120**, and so does the power cables **132**. The strain relief member **114** may also cooperate with the second housing **112** to clamp the high-speed data cables **131** and/or the power cables **132** tightly, such that the high-speed data cables **131** and/or the power cables **132** are protected from separating from the circuit board **120** when they are subjected to excessive pulling forces.

[0168] In some embodiments, the second housing **112** may be embedded with a nut, and a fastener passes through the strain relief member **114** and is threaded to the nut to secure the strain relief

member **114** to the second housing **112**. Where the second housing is made of thermoplastics, the nut may be a knurled nut, embedded into the second housing **112** by heating. The nut may also be secured into a recess reserved in the second housing **112**, for example by adhesives, the second housing **112** may be over-molded on the nut. In this way, the nut may be enclosed and unlikely to fall during disassembly and/or assembly, and the appearance is cleaner. The process can be simplified and the nut is firmly fixed to the second housing **112**. The nut may be made of a metallic material to avoid problems, such as stripped thread.

[0169] According to another aspect of the present disclosure, a second electrical connector **200** is provided. The second electrical connector **200** may, for example, be mated to the first electrical connector **100** described above. The second electrical connector **200** is used to establish an electrical connection between a to-be-mounted circuit board (not shown) and a mating electrical connector. As shown in FIGS. **11** to **14**, the second electrical connector **200** may be configured as a socket connector. For example, the to-be-mounted circuit board may be called as a first circuit board (which may also be referred to as a “first printed circuit board” or “first PCB”), and the mating first electrical connector **100** may be a plug connector. The second electrical connector **200** may be mounted to the to-be-mounted circuit board. Conductive members of the plug connector may be inserted into the second electrical connector **200**, such that the to-be-mounted circuit board may be interconnected with the plug connector via the second electrical connector **200**.

[0170] Optionally, the second electrical connector **200** may also be mated with another electrical device, which may be a second circuit board (which may also be referred to as a “second printed circuit board” or “second PCB”). In this case, the first circuit board may be a motherboard, and the second circuit board may be a daughter card, such as solid state disk (SSD), wireless communication module, RF module. The electrical connector may be mounted to the first circuit board, and conductive pads on the second circuit board may be inserted into the electrical connector, such that the first circuit board and the second circuit board may be interconnected via the electrical connector.

[0171] As shown in FIGS. **13A** and **13B**, the second electrical connector **200** may comprise a main housing **210**, which include a second mating portion **211** for mating with the mating electrical connector. The main housing **210** may be molded with an insulating material, such as plastic. Plastic may include but not limited to liquid crystal polymer (LCP), polyphenylene sulfite (PPS), high-temperature nylon or poly-p-phenylene oxide (PPO), or polypropylene (PP), or other materials may be used. The second electrical connector **200** is configured as a socket connector. For clear and concise description, the mating direction X-X, the first lateral direction Y-Y and the second lateral direction Z-Z are in use hereinafter.

[0172] The second electrical connector **200** may include a plurality of conductive terminals **220**. The plurality of conductive terminals **220** may be held by the main housing **210** and extend from the second mating portion **211** to a second mounting portion **212**. The plurality of conductive terminals **220** each may include a mating end **221**, a mounting end **222** opposite the mating end **221**, and an intermediate portion **223** extending between the mating end **221** and the mounting end **222**. The plurality of conductive terminals **220** may be formed of conductive materials including metal or metal alloy, such as copper or a copper alloy. The mating end **221** may be configured for mating with a corresponding conductive portion of an electrical device such as the aforementioned second circuit board or plug connector, and the mounting end **222** may be configured to be mounted to a circuit board such as the aforementioned first circuit board. The mounting end **222** may be straight along the second lateral direction Z-Z. For example, the first circuit board may include conductive portions such as conductive pads or conductive through-holes, and the mounting ends **222** of the conductive terminals **220** may be configured to be connected to the conductive portions of the first circuit board by any suitable process known in the art (e.g., press-fit or soldering). For example, the second circuit board may include conductive portions disposed at or near an edge thereof and the conductive portions may be inserted into the electrical connector, such

that the conductive portions of the second circuit board electrically contact with the mating ends **221** of the conductive terminals **220**. In this manner, the conductive terminals **220** can establish electrical connections between the first circuit board and the second circuit board, to transmit signal and/or power. As another example, the plug connector may include a circuit board, such as the circuit board **120**, disposed in the first mating portion, and the pads of the circuit board are in contact with the mating ends **221** of the conductive terminals **220**. In this manner, the conductive terminals **220** can establish electrical connections between the first circuit board and the plug connector, to transmit signal and/or power.

[0173] As illustrated in FIG. **13A**, the intermediate portion **223** of each of the plurality of conductive terminals **220** may include a curved portion, such that the mounting end **222** and the mating end **221** of the conductive terminal **220** are oriented substantially perpendicular to each other. The mounting ends **222** are oriented along the second lateral direction Z-Z, and the mating ends **221** are oriented along the mating direction X-X. With this configuration, the second electrical connector **200** can be mounted to the first circuit board perpendicular to the second lateral direction Z-Z and be mated with the plug connector or the second circuit board along the mating direction X-X so that the first circuit board is electrically connected to the plug connector or the second circuit board.

[0174] During the mounting of the second electrical connector **200** onto the first circuit board, and/or during the insertion/detachment of the plug connector into/from the second electrical connector **200** which is already mounted onto the first circuit board, it is prone to occur that parts of the intermediate portions **223** of the conductive terminals **220** adjacent to the mounting ends **222** are bent or even broken, since the intermediate portions **223** of the conductive terminals **220** are curved. In order to avoid such problems, the main housing **210** may include a second mounting portion **212**. The second mounting portion **212** may be secured to the second mating portion **211**, as shown in FIG. **13A**. The second mounting portion **212** may include a protrusion **2121** and a latch **2122**. The protrusion **2121** may limit relative movement direction between the second mounting portion **212** and the second mating portion **211**, and the latch **2122** may determine the position of the second mounting portion **212** relative to the second mating portion **211**. After the second mounting portion **212** is secured to the second mating portion **211**, the second mounting portion **212** may surround and hold at least parts of the intermediate portions **223** between curved portions **223A** and the mounting ends **222**.

[0175] With this configuration, when the second electrical connector **200** is mounted to the first circuit board and/or when the plug connector is inserted into/detached from the second electrical connector **200** which is already mounted onto the first circuit board, the parts of the intermediate portions **223** adjacent to the mounting ends **222** (e.g., parts of the intermediate portions between the curved portions **223A** and the mounting ends **222**) cannot tilt relative to the second lateral direction Z-Z, such that the parts of the intermediate portions **223** adjacent to the mounting ends **222** can be efficiently avoided from bending or even breaking. The second electrical connector **200** has a higher mechanical reliability. Moreover, the second mounting portion **212** is also capable of keeping the conductive terminals **220** in place relative to the main housing **210**.

[0176] The mating end **221** of each conductive terminal may be configured to be inserted into the second mating portion **211** from a rear side along the mating direction X-X. At least a part of the intermediate portion **223** between the mating end **221** and the curved portion **223A** may be held by the second mating portion **211** to keep the conductive terminal in place relative to the insulating housing. For example, at least the part of the intermediate portion **223** between the mating end **221** and the curved portion **223A** may include a barb, for engaging with a wall of a channel formed in the second mating portion **211**. The second mating portion **211** may include a groove **211A** recessed from a surface of the second mating portion **211** facing the mating first electrical connector **100** along the mating direction X-X. The groove **211A** is configured to receive conductive portions of the electrical device. The mating ends **221** of the plurality of conductive terminals **220** are exposed

in the groove **211A** to electrically connect with the conductive portions of the electrical device respectively when the conductive portions are inserted into the groove **211A**. The conductive terminals **220** are able to be in place relative to the main housing **210** by the second mounting portion **212** surrounding and holding at least parts of the intermediate portions **223** of the conductive terminals **220** between the curved portions **223A** and the mounting ends **222** as well as by the second mating portion **211** holding at least parts of the intermediate portions **223** between the mating ends **221** and the curved portions **223A**. It should be appreciated that the present application is not limited thereto.

[0177] In some embodiments, the second mating portion **211** is substantially perpendicular to the second mounting portion **212**, since the mating ends **221** of the conductive terminals **220** held by the second mating portion **211** are substantially perpendicular to the mounting ends **222** held by second mounting portion **212**. It facilitates the assembly of the conductive terminals **220** to the second mating portion **211** and the second mounting portion **212**, as well as provides support for different portions of the conductive terminals **220**.

[0178] The second electrical connector **200** may include a cage **230**, as shown in FIGS. **11A** and **11B**. The cage **230** surrounds outside the main housing **210**, and the cage **230** has a front portion **234** extending beyond the second mating portion **211**. The longer cage **230** may provide better support for the first electrical connector **100** when mated with the first electrical connector **100**. Compared to the main housing **210** or the housing assembly **110** of the mating first electrical connector, the cage **230** may withstand greater external forces, which reduces the risk of breaking the mating portions of the first electrical connector **100** and the second electrical connector **200** by the external forces when the first electrical connector **100** is mated with the second electrical connector **200**. The longer cage **230** may provide a space for mounting an outer housing **240**. Optionally, the cage **230** may be secured to the first circuit board by adhesives, soldering, snaps, etc., to support and position the main housing **210**. The cage **230** may be connected to a signal ground to form an effective shielding against external interference. The second electrical connector **200** may include the outer housing **240** attached to the front portion **234** of the cage **230**. The outer housing **240** may include a complementary locking feature, such as a first complementary locking feature **251** and/or a second complementary locking feature **252**. The complementary locking feature is configured to receive the locking feature of the mating electrical connector, such as the first locking feature **141** and/or the second locking feature **142**. In the embodiment shown in FIG. **14**, the outer housing **240** is complex in shape and has different thicknesses at different regions. Thus, the outer housing **240** may be made by a process such as injection molding. Material suitable for the outer housing **240** may include but not limited to liquid crystal polymer (LCP), polyphenylene sulfite (PPS), high-temperature nylon or poly-p-phenylene oxide (PPO), or polypropylene (PP).

[0179] In some embodiments, the cage **230** may be formed by a metal sheet. Opposite edges of the metal sheet are formed with a mortise and tenon for jointing the edges of the metal sheet. As shown in FIG. **12B**, the cage **230** substantially wraps up the main housing **210** of the second electrical connector **200**, the metal sheet may be stamped into a suitable shape, and the main housing **210** assembled with the conductive terminals **220** is placed into the half-finished stamped cage **230**. Subsequently, the main housing **210** is wrapped up by bending the half-finished stamped cage **230**. In some embodiments, the mortise and tenon **231** is located on a lower portion of the cage **230**. The mortise and tenon **231** can withstand forces parallel to the metal sheet. Compared to soldering the metal sheet to form the cage **230**, costs are lower in mass production of the cage **230** by use of the mortise and tenon **231** and stamping process, and the reliability and yield are higher.

[0180] In some embodiments, the front portion **234** of the cage **230** may include barbs **232** outwardly protruding, and the outer housing **240** is provided with openings **242** for engaging with the barbs **232**. To facilitate the assembly, the barbs **232** may be stamped out of the surface of the cage **230**. After the cage **230** is assembled with the main housing **210**, the outer housing **240** may

be sleeved over the cage **230**. At least one of the outer housing **240** and the barbs **232** may be deformable so that the barbs **232** can be embedded into the openings **242**. After the second electrical connector **200** and the mating electrical connector are connected, the outer housing **240** and the mating electrical connector are interlocked. The outer housing **240** is probably subjected to a pulling force along the mating direction X-X toward the mating electrical connector in use. The barbs **232** of the cage **230** may be hooked to the openings **242** of the outer housing **240** to reduce the risk of the outer housing **240** falling off the cage **230**. The positioning in directions perpendicular to the mating direction X-X can be realized by an inner cavity of the outer housing **240** mated with an external profile of the front portion **234** of the cage **230**.

[0181] In some embodiments, the cage **230** may include a board lock **233**, disposed outside the outer housing **240**. As described above, the cage **230** may be secured to the first circuit board. Since the second electrical connector **200** may withstand a pulling force after being connected to the mating electrical connector, there are certain requirements for the connection strength between the cage **230**, which may be a primary load-bearing component, and the first circuit board, the cage **230** may be connected to the first circuit board by soldering. In some embodiments, one side of the cage **230** may be reliably connected to the first circuit board. In a preferred embodiment, the board lock **233** is protruded from the metal sheet for the cage **230**. The board lock **233** may be inserted in a pad via and/or through-hole of the first circuit board, and may also be soldered to the first circuit board such that the cage **230** is firmly secured to the first circuit board. As shown in FIG. **12B**, the board lock **233** may include first protrusions **233A** and second protrusions **233B**. The first protrusions **233A** may be inserted in reserved through-holes in the first circuit board, to pre-position the cage **230**. The first protrusions **233A** may match the through-holes reserved in the first circuit board so that the cage **230** may not move under slight external forces. This enables the cage **230** to be un-skewed due to swing or slight collision during soldering process. The second protrusions **233B** may match with the pad vias of the first circuit board, and the pad vias are usually slightly larger than the second protrusions **233B**. The second protrusions **233B** are inserted into the pad vias of the first circuit board and fixed reliably to the first circuit board by soldering tin that filling up the gap between the vias and the second protrusions **233B**. The cage **230** is locally heated at the soldering parts by use of the board lock **233**, such that the difficulty of soldering is reduced. Moreover, when the cage **230** is subjected to external forces, the external forces not only act on the soldering parts, but are also dispersed on the substrate of the first circuit board through the board lock **233** penetrating the first circuit board. The pads at the soldering parts are avoided from separating from the substrate of the first circuit board due to the external forces, so as to enhance the firmness of the second electrical connector **200**. Free ends of the second protrusions **233B** may have reduced dimensions, such that steps are formed on sides of the second protrusions **233B** that are nearly flush with a lower surface of the cage **230**. It may allow the reduced ends of the second protrusions **233B** to be inserted into the pad vias of the first circuit board, and the steps may abut against the surface of the first circuit board without entering the pad vias. In this way, the cage is positioned and the cage **230** is unlikely to skew.

[0182] In some embodiments, an inner surface of the cage **230** and an outer surface of the second mating portion **211** are spaced apart from to form a gap **260** for receiving the first mating portion **110A** of the mating electrical connector. In some embodiments, as shown in FIG. **13A**, the second mating portion **211** includes a second mating end **211B** adapted to the first mating portion **110A**, and a second mounting end **211C** engaged with the second mounting portion **212**. The second mounting end **211C** is adapted to the cage **230** in dimensions, and the second mating end **211B** is smaller than the second mounting end **211C** in dimensions. The cage **230** may be fixed to the second mounting end **211C** by structures, such as snaps, while the portion of the cage **230** that accommodates the second mating portion **211B** forms the gap **260** with the second mating portion **211B**. The gap **260** matches with the first mating portion of the mating electrical connector in dimensions. When the second mating portion **211** is connected to the first mating portion **110A** of

the mating electrical connector, the first mating portion **110A** may be inserted into the cage **230**, and the second mating end **211B** of the second mating portion **211** is inserted into the opening of the first mating portion **110A**. The cage **230** may also be provided with resilient sheets **235**, as shown in FIGS. **11A-11B** and **12A-12B**. After the second electrical connector **200** is mated to the first electrical connector **100**, the resilient sheets **235** may press-fit on the outer surface of the first mating portion **110A** of the first electrical connector **100**, reducing the swing of the first mating portion **110A** and/or the second mating portion **211**.

[0183] In some embodiments, the outer housing **240** may be sleeved over the front portion **234** of the cage **230**, and the outer housing **240** includes a locking portion **241** extending beyond the front portion **234**. The locking portion **241** may be provided with a complementary locking feature, which may include a first complementary locking feature **251** and/or a second complementary locking feature **252** such as projection, recess and opening. Optionally, the mating electrical connector may include the first locking feature **141**, and the first complementary locking feature **251** may be engaged with the first locking feature **141** to lock the mating electrical connector with the outer housing **240**, such that the mating electrical connector and the electrical connector **200** do not disconnect accidentally. Optionally, the mating electrical connector may include the second locking feature **142**, and the second locking feature **142** may be engaged with the second complementary locking feature **252** to lock the electrical connector **200** with the mating electrical connector. Optionally, the mating electrical connector may include the first locking feature **141** and the second locking feature **142**, and the outer housing **240** may include the first complementary locking feature **251** and the second complementary locking feature **252** correspondingly. In some embodiments, when the first locking feature **141** is engaged with the first complementary locking feature **251**, the second locking feature **142** is also engaged with the second complementary locking feature **252**. In this case, when either of the first locking feature **141** and the second locking feature **142** is unlocked, the mating electrical connectors can still not be moved. For example, if the first locking feature **141** and the second locking feature **142** both are brought to their respective unlocked positions, the mating electrical connectors can be moved. In another embodiment, when the first locking feature **141** is engaged with the first complementary locking feature **251** during the first electrical connector **100** being mated to the second electrical connector **200**, the mating electrical connector can be moved along the mating direction for a certain distance, until the second locking feature **142** is engaged with the second complementary locking feature **252**.

Correspondingly, during the first electrical connector **100** being unmated to the second electrical connector **200**, after the second locking feature **142** is disengaged with the second complementary locking feature **252**, the mating electrical connector is initially detached from the electrical connector such that the mating electrical connector can be moved for a certain distance. Moreover, the second locking feature **142** remains in the second unlocked position during the partial mating/unmating. The first locking feature **141** can provide the locking between the electrical connector and the mating electrical connector. The mating electrical connector can be completely detached from the electrical connector by simply operating the first locking feature **141** to the first unlocked position. The position of the mating electrical connector partially mated/unmated from the electrical connector can be changed by adjusting the positions of the first complementary locking feature **251** and the second complementary locking feature **252** on the outer housing **240**, or the positions of the first locking feature **141** and the second locking feature **142** on the mating electrical connector.

[0184] In some embodiments, the outer housing **240** has a first dimension in the first lateral direction Y-Y and a second dimension in the second lateral direction Z-Z. The first dimension is greater than the second dimension. The outer housing **240** has first walls **2401** opposite in the first lateral direction Y-Y and second walls **2402** opposite in the second lateral direction Z-Z, as shown in FIG. **14**. One of the second walls **2402** includes a bulge **2402A** outwardly protruding. The complementary locking feature includes first openings (e.g., first complementary locking features

251) in the first walls **2401** and a second opening (e.g., second complementary locking feature **252**) in the bulge **2402A**. Referring back to FIG. **1E**, in the illustrated example, the mating electrical connector includes a protrusion in the second lateral direction Z-Z for accommodating a portion of the second locking feature **142**. Correspondingly, the outer housing **240** may be provided with the bulge **2402A** outwardly protruding from the second wall **2402**. The bulge **2402A** can accommodate the protrusion of the mating electrical connector inside, which thus may be mated with the second locking feature **142** of the mating electrical connector.

[0185] The present disclosure has been described by the above embodiments, but it should be understood that a variety of variations, modifications and improvements may be made according to the teaching of the present disclosure by those skilled in the art, and all of these variations, modifications and improvements fall within the spirit and the scope of protection of the present disclosure. The scope of protection of the present disclosure is defined by the appended claims and its equivalent scope. The above embodiments are only for the purpose of illustration and description, and are not intended to limit the present disclosure to the scope of the described embodiments.

[0186] Moreover, although many creative aspects of the electronic system have been described above with reference to a board connector and a cable connector, it should be understood that the aspects of the present disclosure are not limited to these. Any one of the creative features, whether alone or combined with one or more other creative features, can also be used for two mating cable connectors or two mating board connectors. Also, the connector can be used as a plug connector or a socket connector. The connector may be an orthogonal connector, a vertical connector, a coplanar connector, or a right-angle connector.

[0187] In the description of the present disclosure, it is to be understood that orientation or positional relationships indicated by orientation words “front”, “rear”, “upper”, “lower”, “left”, “right”, “lateral direction”, “mating direction”, “perpendicular direction”, “perpendicular”, “horizontal”, “top”, “bottom” and the like usually are shown based on the accompanying drawings, only for the purposes of the case in describing the present disclosure and simplification of its descriptions. Unless stated to the contrary, these orientation words do not indicate or imply that the specified apparatus or element has to be specifically located, and structured and operated in a specific direction, and therefore, should not be understood as limitations to the present disclosure. The orientation words “inside” and “outside” refer to the inside and outside relative to the contour of each component itself.

[0188] For facilitating description, the spatial relative terms such as “on”, “above”, “on an upper surface of” and “upper” may be used here to describe a spatial position relationship between one or more components or features and other components or features shown in the accompanying drawings. It should be understood that the spatial relative terms not only include the orientations of the components shown in the accompanying drawings, but also include different orientations in use or operation. For example, if the component in the accompanying drawings is turned upside down completely, the component “above other components or features” or “on other components or features” will include the case where the component is “below other components or features” or “under other components or features”. Thus, the exemplary term “above” can encompass both the orientations of “above” and “below”. In addition, these components or features may be otherwise oriented (for example rotated by 90 degrees or other angles) and the present disclosure is intended to include all these cases.

[0189] It should be noted that the terms used herein are only for describing specific embodiments, and are not intended to limit the exemplary embodiments according to the present application. As used herein, an expression of a singular form includes an expression of a plural form unless otherwise indicated. In addition, it should also be understood that when the terms “including” and/or “comprising” are used herein, it indicates the presence of features, steps, operations, parts, components and/or combinations thereof.

[0190] It should be noted that the terms “first”, “second” and the like in the description and claims, as well as the above accompanying drawings, of the present disclosure are used to distinguish similar objects, but not necessarily used to describe a specific order or precedence order. It should be understood that ordinal numbers used in this way can be interchanged as appropriate, so that the embodiments of the present disclosure described herein can be implemented in a sequence other than those illustrated or described herein.

Claims

1. An electrical connector comprising: a housing assembly; a module held by the housing assembly and comprising a plurality of conductive pads along a mating edge disposed in a pitch less than or equal to 1.0 mm and a plurality of conductors extending away from the mating edge; and a connector position assurance (CPA) device coupled to the housing assembly and configured for securing the electrical connector to a mating electrical connector.
2. The electrical connector of claim 1, wherein: the plurality of conductive pads are a plurality of first conductive pads configured for signal transmission; and the module comprises a second conductive pad configured for power transmission.
3. The electrical connector of claim 2, wherein the module comprises: a circuit board disposed inside the housing assembly and comprising the plurality of first conductive pads and the second conductive pad, each of the plurality of first conductive pads having a front end configured to contact a mating electrical connector and a rear end, the second conductive pad having a front end configured to contact the mating electrical connector and a rear end; a plurality of first cables comprising first wires disposed in a first pitch less than or equal to 1.0 mm, the first wires having ends mounted to the rear ends of the plurality of first conductive pads of the circuit board so as to electrically connect to the plurality of conductive pads; and a plurality of second cables comprising second wires disposed in a second pitch greater than the first pitch, the second wires comprising ends mounted to the rear end of the second conductive pad.
4. The electrical connector of claim 1, wherein: the plurality of conductive pads conform to Peripheral Component Interconnect Express Generation 5 and/or Generation 6 for transmission rates.
5. The electrical connector of claim 1, wherein: the CPA device comprises a first locking feature having a first locked position and a first unlocked position, and a second locking feature having a second locked position and a second unlocked position; the first locking feature and the second locking feature are disposed on different sides of the housing assembly; and the first locking feature is configured to be movable from the first locked position to the first unlocked position when the second locking feature is in the first unlocked position and to remain in the first locked position when the second locking feature is in the second locked position.
6. The electrical connector of claim 5, wherein: the CPA device comprises a third locking features having a third locked position and a third unlocked position; and the third locking feature is coupled to the second locking feature such that the second locking feature is in the second locked position when the third locking feature is in the third locked position and the second locking feature is in the second unlocked position when the third locking feature is in the third unlocked position.
7. The electrical connector of claim 5, wherein: the first locking feature comprises a spring biased element; and the second locking feature comprises a pull tab.
8. An electrical connector comprising: a housing assembly; a module held by the housing assembly and comprising a plurality of conductive pads; and a connector position assurance (CPA) device coupled to the housing assembly and configured for securing the electrical connector to a mating electrical connector, the CPA device comprising first and second locking features disposed on different sides of the housing assembly.

- 9.** The electrical connector of claim 8, wherein: the CPA device comprises a first locking feature having a first locked position and a first unlocked position, and a second locking feature having a second locked position and a second unlocked position; and the first locking feature is configured to be movable from the first locked position to the first unlocked position when the second locking feature is in the first unlocked position and to remain in the first locked position when the second locking feature is in the second locked position.
- 10.** The electrical connector of claim 9, wherein: the CPA device comprises a third locking features having a third locked position and a third unlocked position; and the third locking feature is coupled to the second locking feature such that the second locking feature is in the second locked position when the third locking feature is in the third locked position and the second locking feature is in the second unlocked position when the third locking feature is in the third unlocked position.
- 11.** The electrical connector of claim 10, wherein the CPA device comprises: a first operating member configured to be operable in a mating direction so as to move the first locking feature between the first locked position and the first unlocked position; and a second operating member configured to be operable in a direction perpendicular to the mating direction so as to move the second locking feature between the second locked position and the second unlocked position.
- 12.** The electrical connector of claim 11, wherein the third locking feature is configured to: engage the first operating member when the third locking feature is in the third locked position so as to restrain movement of the first operating member, and disengage the first operating member when the third locked feature is in the third unlocked position.
- 13.** The electrical connector of claim 11, wherein the first locking feature comprises: a first locking head configured to engage a first side of the mating electrical connector; a first tail configured to engage the first operating member; and an intermediate portion joining the first locking head and the first tail and pivotably connected to the housing assembly about a first axis such that the first locking feature is pivotable between the first locked position and the first unlocked position.
- 14.** The electrical connector of claim 13, wherein: the first locking feature comprises a resilient beam configured to abut the housing assembly so as to bias the first locking head toward outside of the housing assembly; and the electrical connector comprises a resilient member disposed between the second locking feature and the housing assembly and configured to bias the second locking feature toward the second locked position.
- 15.** The electrical connector of claim 13, wherein: the second operating member is configured to engage a second side of the mating electrical connector; and the second operating member comprises the first locking feature, the second locking feature, and an intermediate portion joining the first locking feature and the second locking feature and pivotably connected to the housing assembly about a second axis perpendicular to the first axis.
- 16.** The electrical connector of claim 9, wherein: the plurality of conductive pads are dimensioned to be in electrical contact with the mating electrical connector as long as the first locking feature is in the first locked position.
- 17.** An electrical connector, comprising: a main housing comprising a mating portion for engaging a mating electrical connector; a plurality of conductive terminals held by the main housing; a cage disposed outside the main housing and comprising a front portion extending beyond the mating portion; and an outer housing attached to the front portion of the cage, the outer housing comprising first and second locking features configured to engage complementary locking features of the mating electrical connector, the first locking feature disposed on a first side of the outer housing and the second locking feature disposed on a second side of the outer housing.
- 18.** The electrical connector of claim 17, wherein: the cage comprises a barb protruding outwardly; and the outer housing comprises an opening for engaging the barb.
- 19.** The electrical connector of claim 17, wherein: the outer housing comprises a locking portion extending beyond the front portion of the cage, the locking portion comprising the first and second

locking features; the locking portion of the outer housing comprises first walls opposite each other and a second wall joining the first walls, the second wall comprising a bulge protruding outwardly; the first locking feature comprises first openings extending through the first walls; and the second locking feature comprises a second opening extending through the bulge.

20. An automotive wiring harness, comprising: the electrical connector of claim 17, wherein the main housing comprises a slot configured to receive a substrate comprising a plurality of conductive pads disposed in a pitch less than or equal to 1.0 mm.
